

Aurora-FV: 2-D Unstructured Compressible Navier–Stokes Benchmark

Submitted solver team

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Abstract

This report is generated from the submitted JSON, CSV, VTK, and manifest files. It documents the finite-volume method, distributed METIS mesh, requested versus observed controls, force histories, and the $Re=200$ transient diagnostic. Terminal steady cases are described as measured plateaus when the requested residual reduction is not reached.

1 Governing equations and nondimensionalization

The conservative state is $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$. The equations solved are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y},$$

with $\mathbf{F}^i = [\rho u, \rho u^2 + p, \rho uv, u(\rho E + p)]^T$ and $\mathbf{G}^i = [\rho v, \rho uv, \rho v^2 + p, v(\rho E + p)]^T$. The calorically perfect closure is $p = (\gamma - 1)\rho e$, $E = e + (u^2 + v^2)/2$, and $a = \sqrt{\gamma p / \rho}$. Reference coefficients use $q_\infty = \rho_\infty U_\infty^2 / 2$ and $C_D, C_L = F_{x,y} / (q_\infty A_{ref})$. Laminar runs use $\mu = \rho_\infty U_\infty L_{ref} / Re$, Newtonian stresses, and $\mathbf{q} = -\mu c_p \nabla T / Pr$.

2 Meshes, boundary conditions, and case parameters

The CGNS reader handles TRI/QUAD zones, geometrically pairs coincident multi-zone faces, computes cell areas and oriented face normals, and retains section names for boundary dispatch. The submitted case table is:

Case	cells	faces	M_∞	Re	mode	boundary tags
cylinder_m010_laminar_re20	10185	20180	0.1	20	laminar	WALL: no_slip_adiabatic_wall; FAR: farfield
cylinder_m010_laminar_re200	10185	20180	0.1	200	laminar	WALL: no_slip_adiabatic_wall; FAR: farfield
naca0012_m015_inviscid	20816	36498	0.15	0	inviscid	bc-2: farfield; bc-4: slip_wall
naca0012_m015_laminar_re5000	20816	36498	0.15	5000	laminar	bc-2: farfield; bc-4: no_slip_adiabatic_wall
naca0012_m080_inviscid	20816	36498	0.8	0	inviscid	bc-2: farfield; bc-4: slip_wall
naca0012_m080_laminar_re5000	20816	36498	0.8	5000	laminar	bc-2: farfield; bc-4: no_slip_adiabatic_wall
naca0012_m200_inviscid	20816	36498	2	0	inviscid	bc-2: farfield; bc-4: slip_wall
naca0012_m200_laminar_re5000	20816	36498	2	5000	laminar	bc-2: farfield; bc-4: no_slip_adiabatic_wall

Farfield faces use the configured freestream state; inviscid walls use a mirrored normal velocity and exact zero mass/energy wall flux; no-slip adiabatic walls use zero wall velocity, pressure traction, and gradient-based viscous traction. Surface rows are boundary values: no-slip $u = v = 0$, while slip walls retain tangential velocity and have zero normal velocity.

3 Spatial discretization and positivity

Each owned cell accumulates oriented face fluxes. The Rusanov local Lax–Friedrichs flux supplies the approximate Riemann solve. For the low-Mach $\text{Re}=200$ transient, an acoustic-scaled all-speed Rusanov signal applies the measured cutoff from metadata only to the artificial acoustic contribution; the conservative physical flux and supplied Rusanov scale remain unchanged. This is not a full pressure–velocity preconditioned flux. Primitive variables are reconstructed with a least-squares gradient, exchanged on the halo, and limited by a Barth–Jespersen bound. Density and pressure are protected by a bounded primitive-variable line search before conservative storage. The same reconstructed gradients provide Newtonian stress and Fourier heat flux; the force postprocessor separates pressure and tangential viscous traction.

4 Implicit and transient integration

The current steady implementation uses scalar Jacobi/Richardson with an MPI-global Armijo backtracking safeguard; the checked-in canonical steady packages predate that safeguard, so their plateau histories are retained as historical evidence. An optional frozen 4x4 block-Jacobi correction is implemented and recorded when explicitly enabled. The requested CFL continuation and the effective cap used by each run are recorded in metadata; conservative caps were selected for the stiff high-Mach/laminar cases. The diagonal update contains the standard $(1 + CFL)^{-1}$ implicit denominator and a conservative positivity line search. The requested and observed controls are:

Case	supplied max	actual	CFL_0	CFL_{max}	inner min	inner max	inner target	observed mean
cylinder_m010_laminar_re20	30000	300	1	100	3	80	0.01	80
cylinder_m010_laminar_re200	–	30000	1	1	5	1000	0.001	67.5
naca0012_m015_inviscid	20000	300	1	100	3	50	0.01	50
naca0012_m015_laminar_re5000	40000	300	1	100	3	80	0.01	80
naca0012_m080_inviscid	30000	300	1	100	3	50	0.01	50
naca0012_m080_laminar_re5000	40000	300	0.5	80	3	80	0.01	80
naca0012_m200_inviscid	40000	300	0.2	50	3	80	0.01	80
naca0012_m200_laminar_re5000	50000	300	0.2	50	3	100	0.01	100

The $\text{Re}=200$ directory uses nominal BDF2 outer steps with $\Delta t = 0.01$ and frozen U^n, U^{n-1} histories in the normal residual; documented converged backward-Euler substeps replace rejected nominal intervals. Histories are updated only after each accepted physical step. Its scalar Jacobi/Richardson inner corrections use the measured reconstruction slope, gain, and all-speed cutoff recorded in metadata. Fresh transient starts in the current source retain the freestream plus an explicit localized symmetry-breaking phase seed and do not apply the synthetic steady wall-profile initializer; no-slip is imposed through the wall flux and gradient treatment. The seed is a one-time initial condition, not persistent forcing. The checked-in 30,000-step package predates the current seed-shape and recovery corrections, so its failed force history is retained as historical production evidence; corrected-seed and corrected-recovery runs are diagnostic-only and are not mixed into the final statistics. The implementation records actual inner minima, maxima, means, target misses, and converged-step fraction, keeps the supplied 1000-iteration allowance as an upper bound, may abandon a clearly stagnant trial before trying a lower pseudo-time gain, and retries a nonlinear stall with converged backward-Euler substeps of 2, 4, 8, 16, 32, 64, or 128 over exactly one nominal interval before rejecting the physical step. After an accepted fallback, the next interval resumes from the accepted state and permits the ordinary BDF2 extrapolated predictor before retrying the normal residual. New transient checkpoints also write an optional U^{n-1} sidecar, so a restart can restore the exact BDF2 pair; legacy state-only checkpoints without that sidecar use one backward-Euler residual to reconstruct the missing history. Restart metadata records usage, resume step, and whether the sidecar was restored; a requested seed may be applied once to a legacy restarted initial state. The fallback interval, substep, recovery-marker, damping-retry, and best-effort continuation counts are recorded in metadata. Any best-effort continuation is diagnostic-only and excludes a final statistically-periodic claim.

5 MPI/METIS decomposition

Rank zero performs one-time CGNS preprocessing and METIS graph partitioning, writes compact rank-local topology, then releases the full mesh. Iterations store owned cells plus one-ring ghosts and exchange only neighbor state/gradient payloads with nonblocking MPI. Residuals and forces use reductions; final fields are gathered only after the solve. The per-rank evidence table (owned, ghost, neighbor, send, and receive counts) is:

Case	rank	owned	ghost	neighbors	send	receive
cylinder_m010_laminar_re20	0	1265	139	6	132	139
cylinder_m010_laminar_re20	1	1289	110	3	115	110
cylinder_m010_laminar_re20	2	1276	115	4	114	115
cylinder_m010_laminar_re20	3	1269	122	6	122	122
cylinder_m010_laminar_re20	4	1291	104	4	105	104
cylinder_m010_laminar_re20	5	1246	140	5	138	140
cylinder_m010_laminar_re20	6	1274	111	4	113	111
cylinder_m010_laminar_re20	7	1275	120	4	122	120
cylinder_m010_laminar_re200	0	2554	135	3	138	135
cylinder_m010_laminar_re200	1	2540	151	3	149	151
cylinder_m010_laminar_re200	2	2541	145	3	145	145
cylinder_m010_laminar_re200	3	2550	110	3	109	110
naca0012_m015_inviscid	0	2593	94	2	94	94
naca0012_m015_inviscid	1	2598	102	4	102	102
naca0012_m015_inviscid	2	2625	90	2	90	90
naca0012_m015_inviscid	3	2648	127	5	127	127
naca0012_m015_inviscid	4	2582	93	2	93	93
naca0012_m015_inviscid	5	2568	98	3	98	98
naca0012_m015_inviscid	6	2592	188	6	188	188
naca0012_m015_inviscid	7	2610	52	2	52	52
naca0012_m015_laminar_re5000	0	2593	94	2	94	94
naca0012_m015_laminar_re5000	1	2598	102	4	102	102
naca0012_m015_laminar_re5000	2	2625	90	2	90	90
naca0012_m015_laminar_re5000	3	2648	127	5	127	127
naca0012_m015_laminar_re5000	4	2582	93	2	93	93
naca0012_m015_laminar_re5000	5	2568	98	3	98	98
naca0012_m015_laminar_re5000	6	2592	188	6	188	188
naca0012_m015_laminar_re5000	7	2610	52	2	52	52
naca0012_m080_inviscid	0	2593	94	2	94	94
naca0012_m080_inviscid	1	2598	102	4	102	102
naca0012_m080_inviscid	2	2625	90	2	90	90
naca0012_m080_inviscid	3	2648	127	5	127	127
naca0012_m080_inviscid	4	2582	93	2	93	93
naca0012_m080_inviscid	5	2568	98	3	98	98
naca0012_m080_inviscid	6	2592	188	6	188	188
naca0012_m080_inviscid	7	2610	52	2	52	52
naca0012_m080_laminar_re5000	0	2593	94	2	94	94
naca0012_m080_laminar_re5000	1	2598	102	4	102	102
naca0012_m080_laminar_re5000	2	2625	90	2	90	90
naca0012_m080_laminar_re5000	3	2648	127	5	127	127
naca0012_m080_laminar_re5000	4	2582	93	2	93	93
naca0012_m080_laminar_re5000	5	2568	98	3	98	98
naca0012_m080_laminar_re5000	6	2592	188	6	188	188
naca0012_m080_laminar_re5000	7	2610	52	2	52	52
naca0012_m200_inviscid	0	2593	94	2	94	94
naca0012_m200_inviscid	1	2598	102	4	102	102
naca0012_m200_inviscid	2	2625	90	2	90	90
naca0012_m200_inviscid	3	2648	127	5	127	127
naca0012_m200_inviscid	4	2582	93	2	93	93
naca0012_m200_inviscid	5	2568	98	3	98	98
naca0012_m200_inviscid	6	2592	188	6	188	188
naca0012_m200_inviscid	7	2610	52	2	52	52
naca0012_m200_laminar_re5000	0	2593	94	2	94	94
naca0012_m200_laminar_re5000	1	2598	102	4	102	102
naca0012_m200_laminar_re5000	2	2625	90	2	90	90
naca0012_m200_laminar_re5000	3	2648	127	5	127	127
naca0012_m200_laminar_re5000	4	2582	93	2	93	93
naca0012_m200_laminar_re5000	5	2568	98	3	98	98

naca0012_m200_laminar_re5000	6	2592	188	6	188	188
naca0012_m200_laminar_re5000	7	2610	52	2	52	52

The aggregate load-balance and communication evidence is:

Case	owned ratio	mean ghosts	max neighbors	total send	total receive
cylinder_m010_laminar_re20	1.036	120.1	6	961	961
cylinder_m010_laminar_re200	1.006	135.2	3	541	541
naca0012_m015_inviscid	1.031	105.5	6	844	844
naca0012_m015_laminar_re5000	1.031	105.5	6	844	844
naca0012_m080_inviscid	1.031	105.5	6	844	844
naca0012_m080_laminar_re5000	1.031	105.5	6	844	844
naca0012_m200_inviscid	1.031	105.5	6	844	844
naca0012_m200_laminar_re5000	1.031	105.5	6	844	844

The independent rank-count runs used the same executable and case inputs. Their measured timings, terminal residuals, and final forces are reported below. The first rank count for a case is the reference; the measured pairwise differences are: naca0012_m015_inviscid: np=2 to np=8 gives $|\Delta C_D| = 0$, $|\Delta C_L| = 2.28e - 10$, $|\Delta R_2| = 4.13e - 13$, and measured speedup 1.54. cylinder_m010_laminar_re20: np=2 to np=8 gives $|\Delta C_D| = 0$, $|\Delta C_L| = 0$, $|\Delta R_2| = 6.5e - 13$, and measured speedup 2.45. The production iteration path uses only neighbor-scoped nonblocking state and gradient exchanges; the final gather is outside the iteration loop.

Case	ranks	wall s	steps	R_2	C_D	C_L	owned	ghost
naca0012_m015_inviscid	2	4.6071	20	0.003994	0.9135	-1.041e-07	10367	117
naca0012_m015_inviscid	8	2.982	20	0.003994	0.9135	-1.043e-07	2593	94
cylinder_m010_laminar_re20	2	113.42	300	1.995	0.1304	-2.667e-09	5088	102
cylinder_m010_laminar_re20	8	46.228	300	1.995	0.1304	-2.667e-09	1265	139

6 Results and force analysis

Case	ranks	steps	time	wall s	residual orders	status / assessment
cylinder_m010_laminar_re20	8	300	0	61.771	0.3432	converged (stable plateau; residual target not reached)
cylinder_m010_laminar_re200	4	30000	300	6060	7.155	failed (unresolved lift oscillation)
naca0012_m015_inviscid	8	300	0	51.277	0.1673	converged (stable plateau; residual target not reached)
naca0012_m015_laminar_re5000	8	300	0	255.72	0.4265	converged (stable plateau; residual target not reached)
naca0012_m080_inviscid	8	300	0	46.792	0.1689	converged (stable plateau; residual target not reached)
naca0012_m080_laminar_re5000	8	300	0	120.48	0.4603	converged (stable plateau; residual target not reached)
naca0012_m200_inviscid	8	300	0	74.09	0.2157	converged (stable plateau; residual target not reached)
naca0012_m200_laminar_re5000	8	300	0	285.68	0.4653	converged (stable plateau; residual target not reached)

Case	mean C_D	mean C_L	final C_m	pressure drag	viscous drag	lift amplitude	convergence note
cylinder_m010_laminar_re20	0.1309	-3.545e-10	-3.341e-10	-0.002028	0.1324	1.349e-09	stable force plateau; residual target not reached
cylinder_m010_laminar_re200	2.816	-9.743e-05	-0.0002278	2.062	0.7541	7.552e-07	unresolved lift oscillation
naca0012_m015_inviscid	0.8817	1.39e-07	-1.345e-06	0.862	0	2.245e-06	stable force plateau; residual target not reached

naca0012_m015_laminar_re5000	0.004801	-1.688e-07	-5.415e-07	-3.302e-08	0.004884	5.953e-07	stable residual reached	force target	plateau; not
naca0012_m080_inviscid	0.202	-8.711e-08	-1.892e-07	0.1968	0	1.944e-07	stable residual reached	force target	plateau; not
naca0012_m080_laminar_re5000	0.00374	-1.577e-08	1.024e-08	-3.088e-09	0.0037	2.781e-08	stable residual reached	force target	plateau; not
naca0012_m200_inviscid	0.1222	1.822e-09	3.127e-09	0.1184	0	2.277e-09	stable residual reached	force target	plateau; not
naca0012_m200_laminar_re5000	0.003666	1.06e-09	1.022e-08	-2.156e-09	0.003523	9.103e-09	stable residual reached	force target	plateau; not

NACA0012 runs are at zero incidence, so the reported lift is near the symmetry level. Inviscid force columns have zero viscous contribution by construction; laminar runs show a positive pressure/skin-friction split. Cylinder Re=20 is assessed from its measured residual and force history. A steady row labelled “stable plateau” has bounded forces but did not meet the requested residual reduction; it is not silently promoted to residual-target convergence.

6.1 Re=200 diagnostic

The Re=200 history has mean $C_D = 2.816$ and lift half-amplitude $7.552e - 07$. The lift signal does not meet the resolved-shedding criterion; no frequency or Strouhal claim is made, and the case remains failed. Measured inner target fraction: 1; target misses: 1; bounded pseudo-time damping retries: 26720 steps (53440 BDF2 attempts, 68 BE-gain retries); bounded backward-Euler fallback intervals: 26720 (53476 substeps, using 2/4/8 substeps as needed); reported BE recovery-marker count: 26720 (this historical counter is not a one-time restart-history count; only the first resumed interval reconstructs missing BDF2 history); best-effort target-miss continuations: 0; restart used: True at accepted step 3280; BDF2 history sidecar restored: False; startup perturbation amplitude: 0.001; Rusanov dissipation scale: 1; all-speed acoustic cutoff: 0.5; reconstruction-slope factor: 0.1; scalar correction gain: 0.4; stationary transient wall start: False.

7 Figures and source traceability

Every generated image is introduced below with a label and a reference; filenames and source paths are recorded in `figure_manifest.csv`. cylinder_m010_laminar_re20: Figure 1, Figure 2, Figure 3, Figure 4, Figure 5, Figure 6, Figure 7. cylinder_m010_laminar_re200: Figure 8, Figure 9, Figure 10, Figure 11, Figure 12, Figure 13, Figure 14, Figure 15, Figure 16, Figure 17, Figure 18. naca0012_m015_inviscid: Figure 19, Figure 20, Figure 21, Figure 22, Figure 23, Figure 24, Figure 25. naca0012_m015_laminar_re5000: Figure 26, Figure 27, Figure 28, Figure 29, Figure 30, Figure 31, Figure 32. naca0012_m080_inviscid: Figure 33, Figure 34, Figure 35, Figure 36, Figure 37, Figure 38, Figure 39. naca0012_m080_laminar_re5000: Figure 40, Figure 41, Figure 42, Figure 43, Figure 44, Figure 45, Figure 46. naca0012_m200_inviscid: Figure 47, Figure 48, Figure 49, Figure 50, Figure 51, Figure 52, Figure 53. naca0012_m200_laminar_re5000: Figure 54, Figure 55, Figure 56, Figure 57, Figure 58, Figure 59, Figure 60.

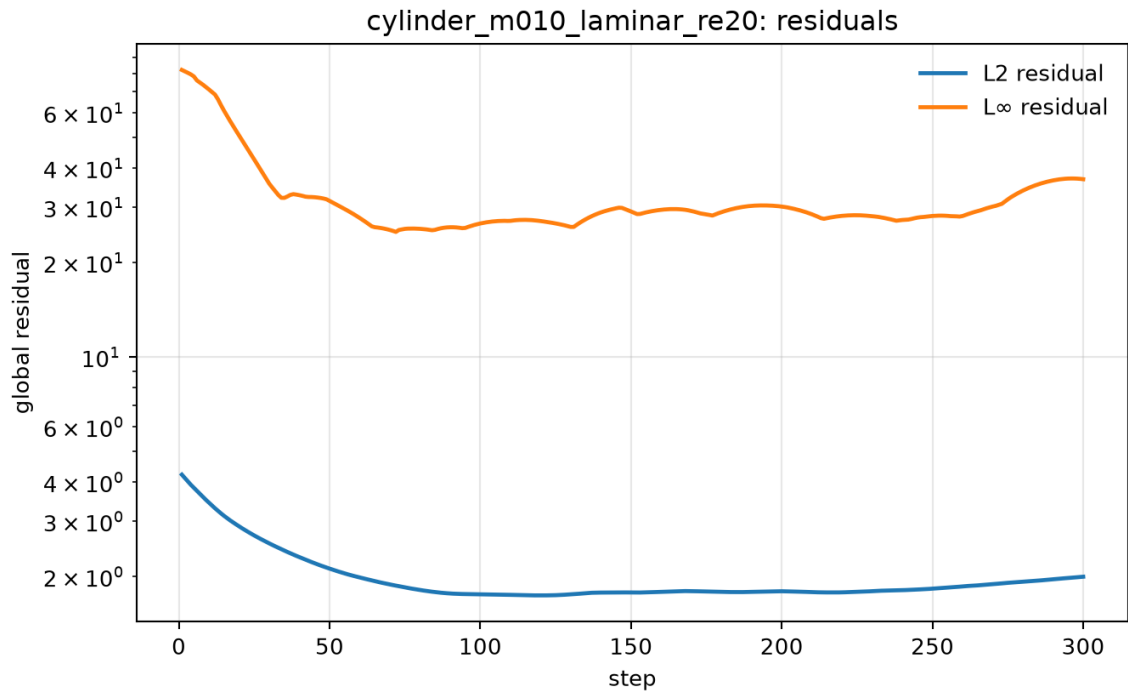


Figure 1: cylinder_m010_laminar_re20: line plot of L2 and L-infinity global residuals (full domain) derived from residuals.csv.

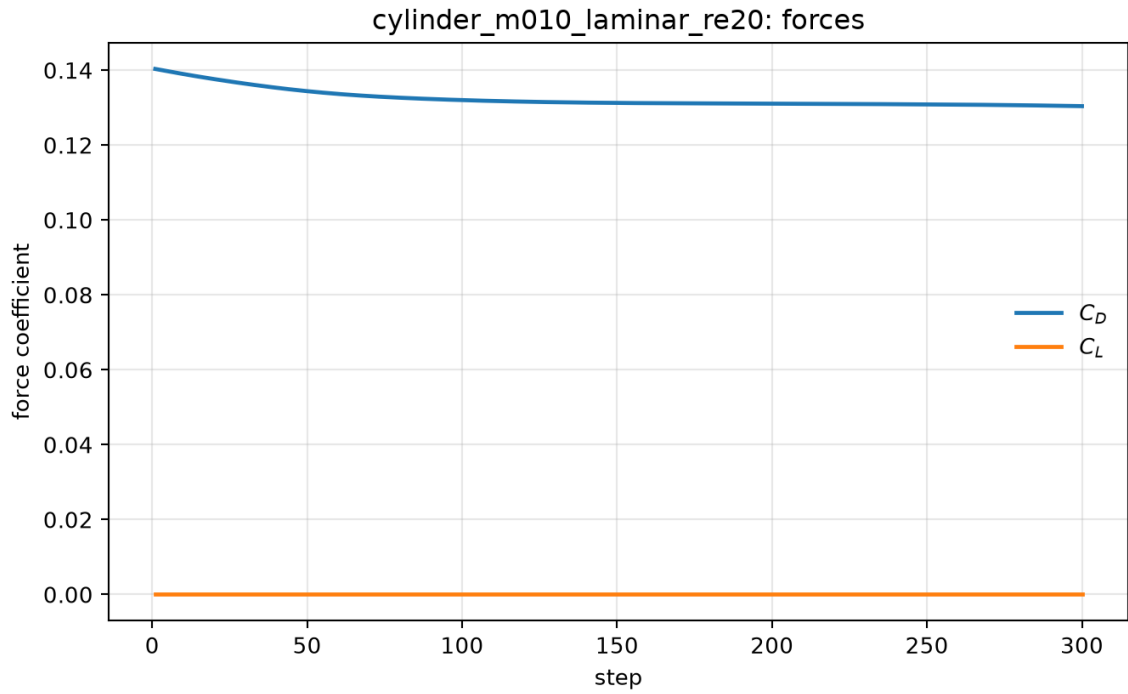


Figure 2: cylinder_m010_laminar_re20: line plot of drag (C_D) and lift (C_L) coefficients (full domain) derived from forces.csv.

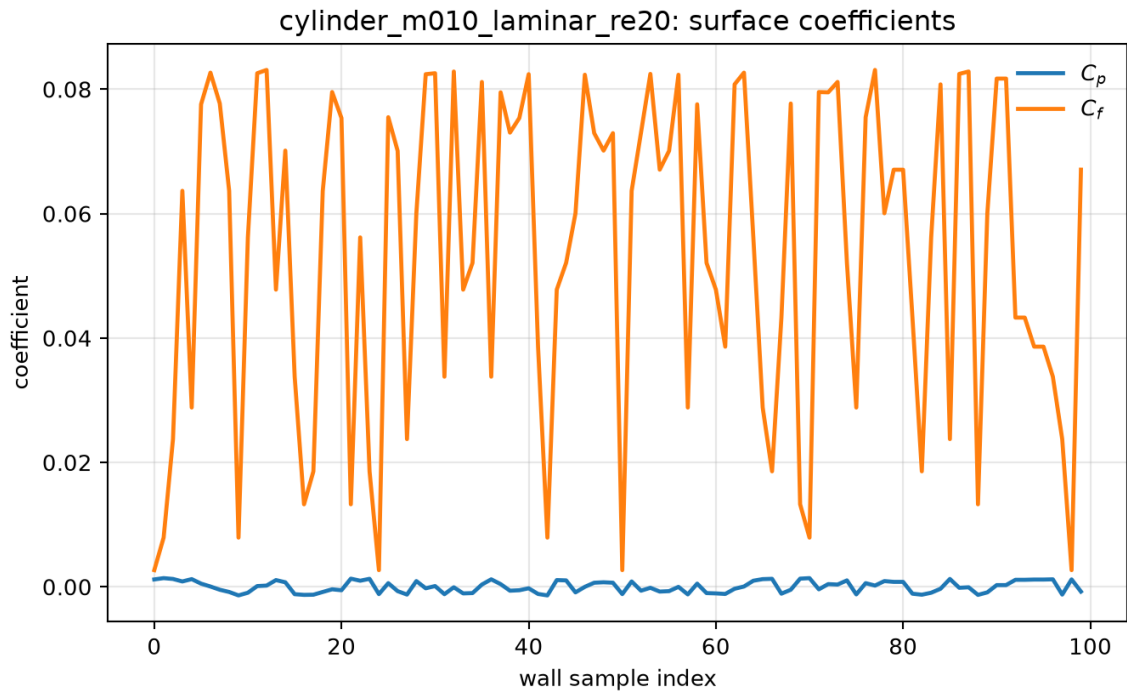


Figure 3: cylinder_m010_laminar_re20: line plot of pressure (C_p) and skin-friction (C_f) surface coefficients (full domain) derived from surface.csv.

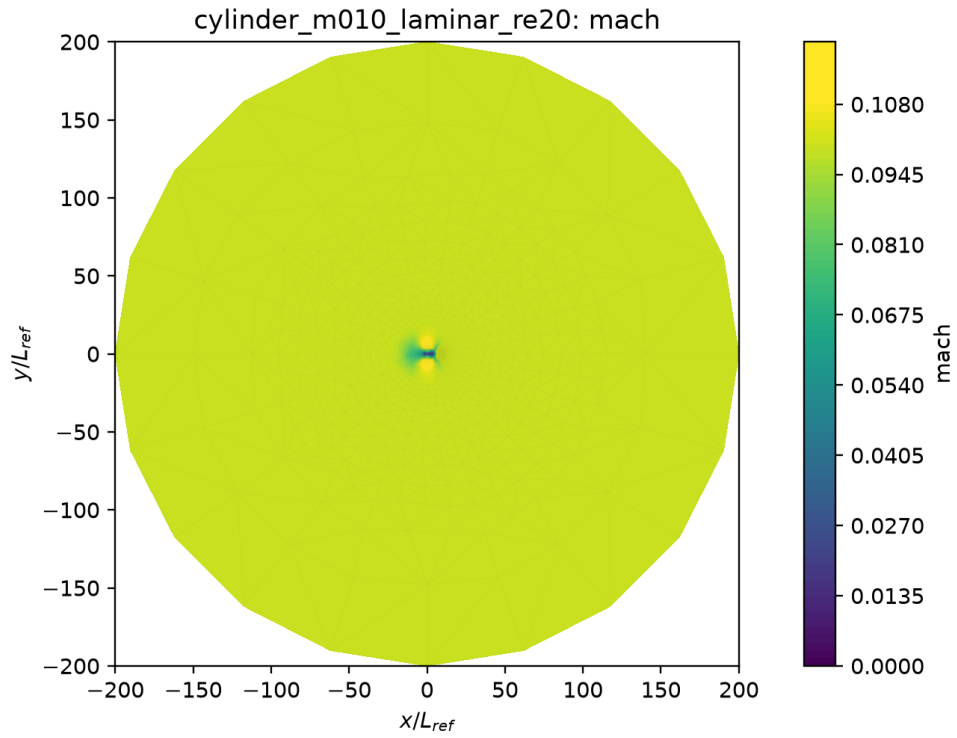


Figure 4: cylinder_m010_laminar_re20: color-filled mach contour (full domain) derived from field_final.vtk.

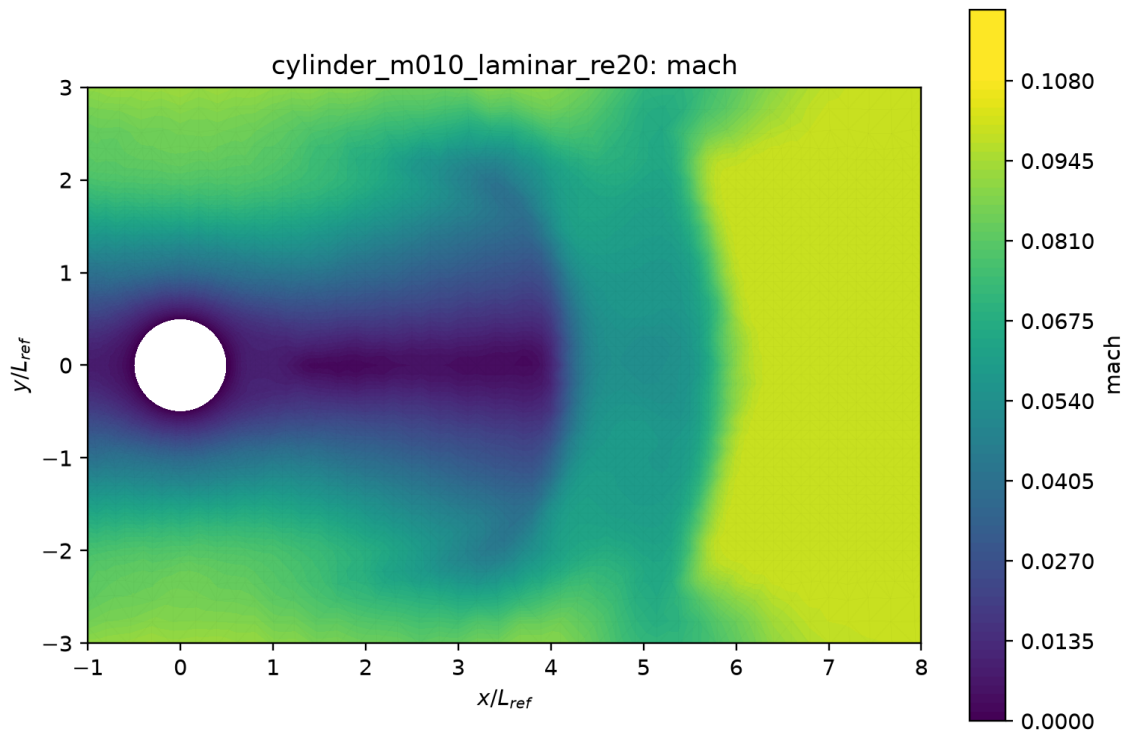


Figure 5: cylinder_m010_laminar_re20: color-filled mach contour (body/wake zoom) derived from field_final.vtk.

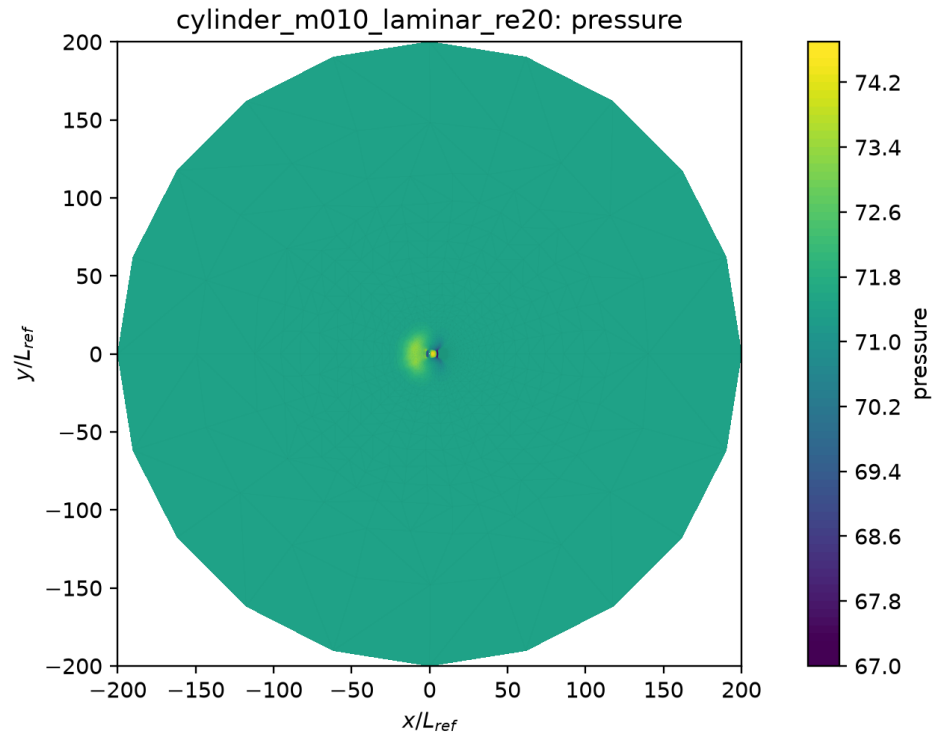


Figure 6: cylinder_m010_laminar_re20: color-filled pressure contour (full domain) derived from field_final.vtk.

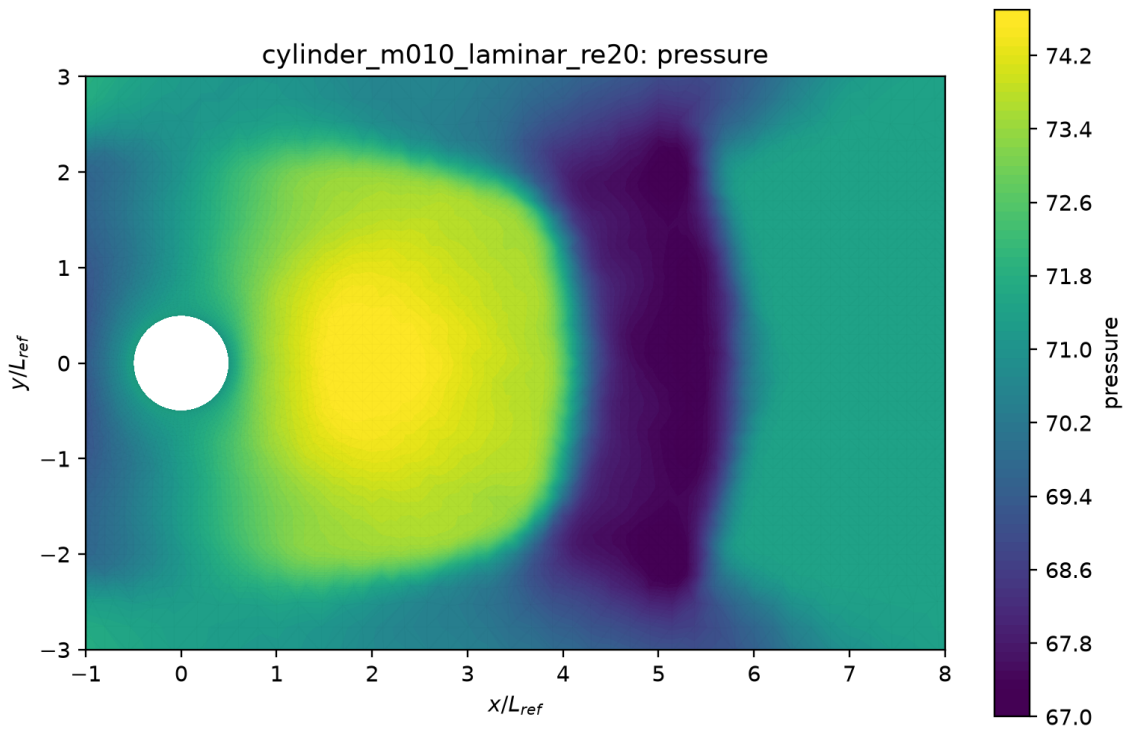


Figure 7: cylinder_m010_laminar_re20: color-filled pressure contour (body/wake zoom) derived from field_final.vtk.

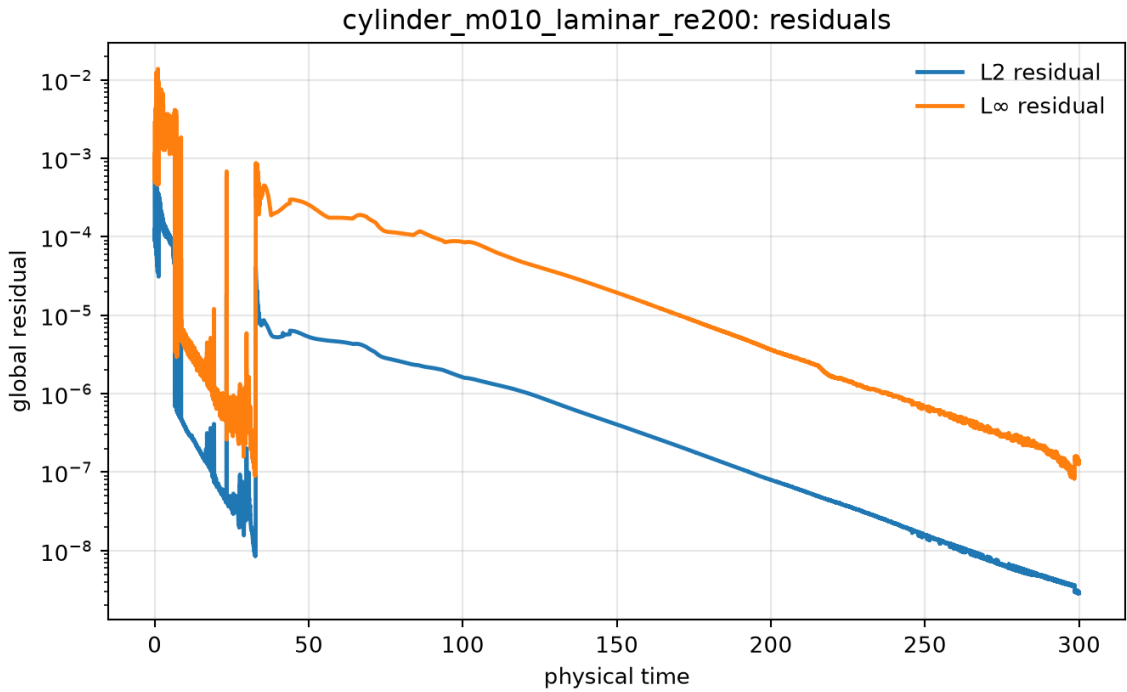


Figure 8: cylinder_m010_laminar_re200: line plot of L2 and L-infinity global residuals (full domain) derived from residuals.csv.

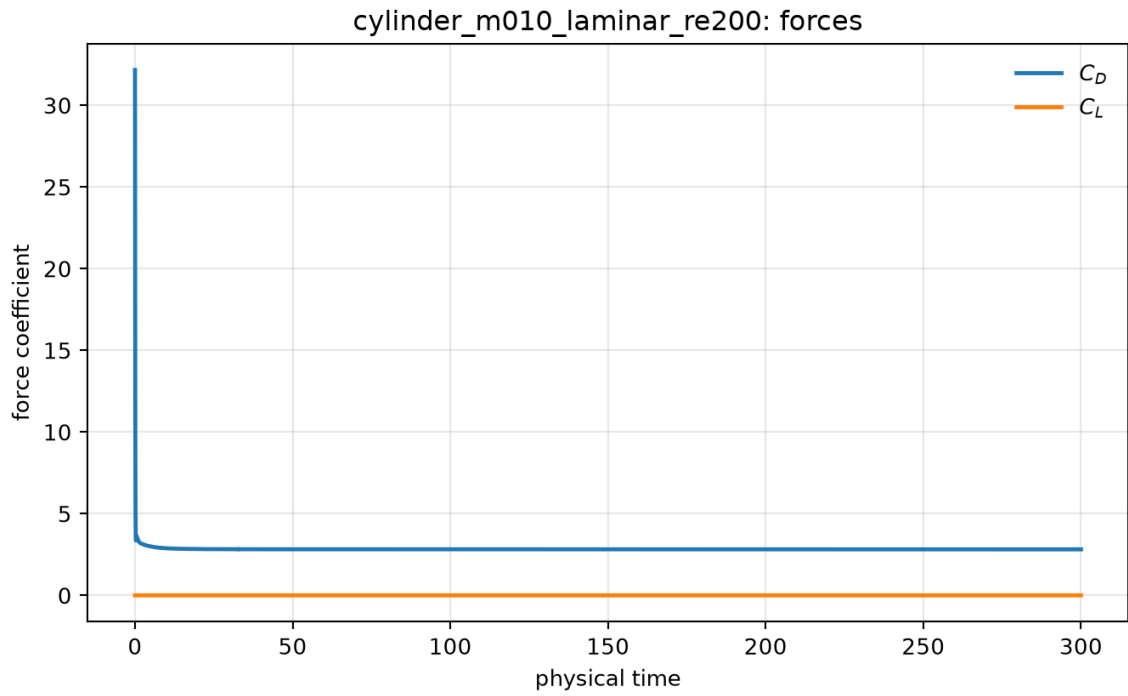


Figure 9: cylinder_m010_laminar_re200: line plot of drag (C_D) and lift (C_L) coefficients (full domain) derived from forces.csv.

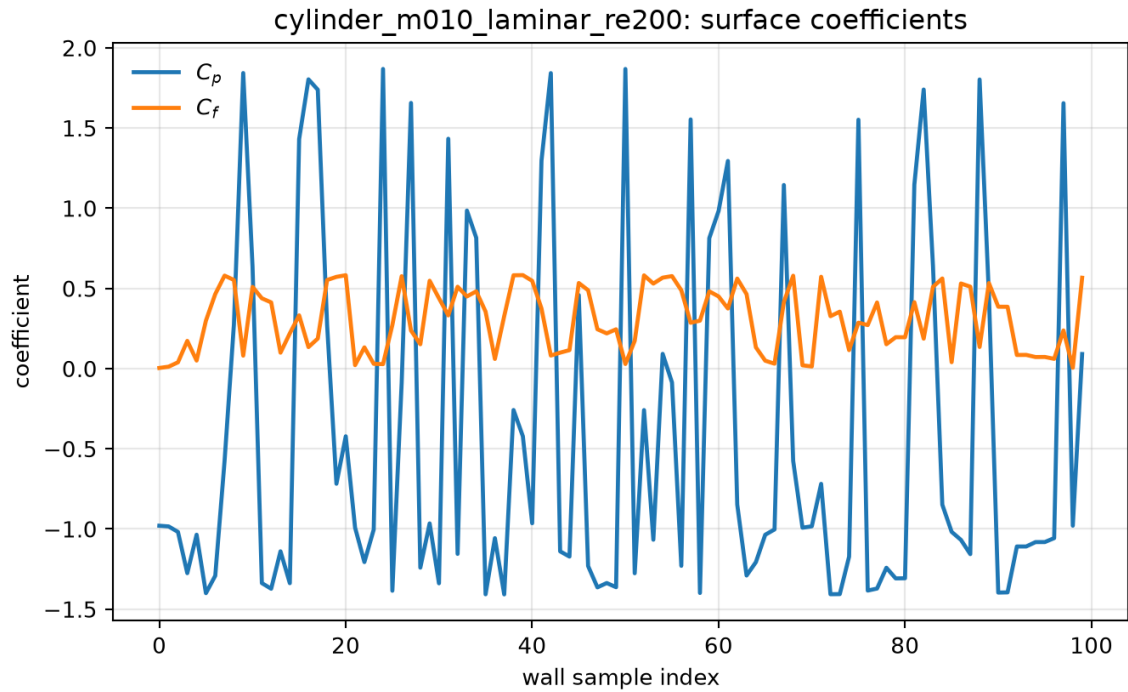


Figure 10: cylinder_m010_laminar_re200: line plot of pressure (C_p) and skin-friction (C_f) surface coefficients (full domain) derived from surface.csv.

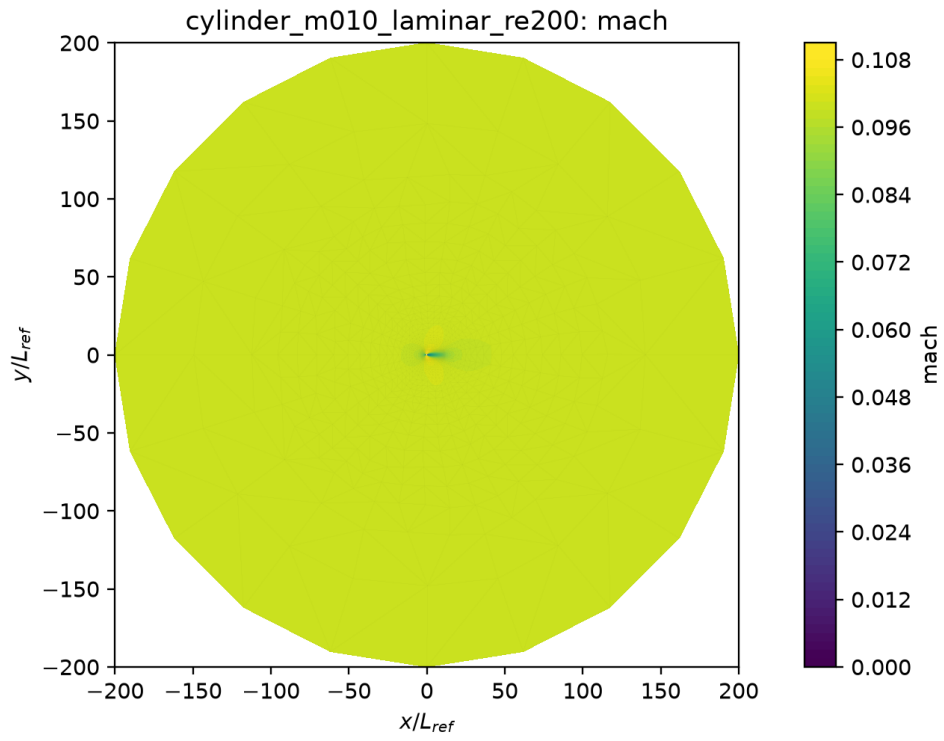


Figure 11: cylinder_m010_laminar_re200: color-filled mach contour (full domain) derived from field_final.vtk.

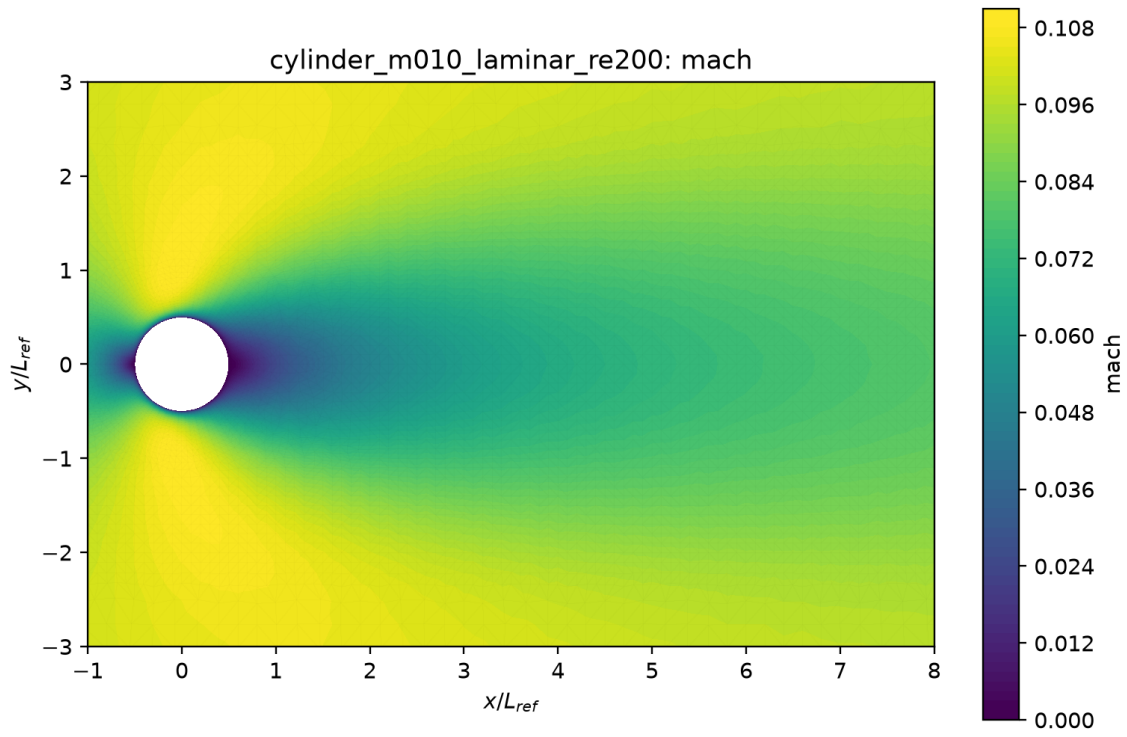


Figure 12: cylinder_m010_laminar_re200: color-filled mach contour (body/wake zoom) derived from field_final.vtk.

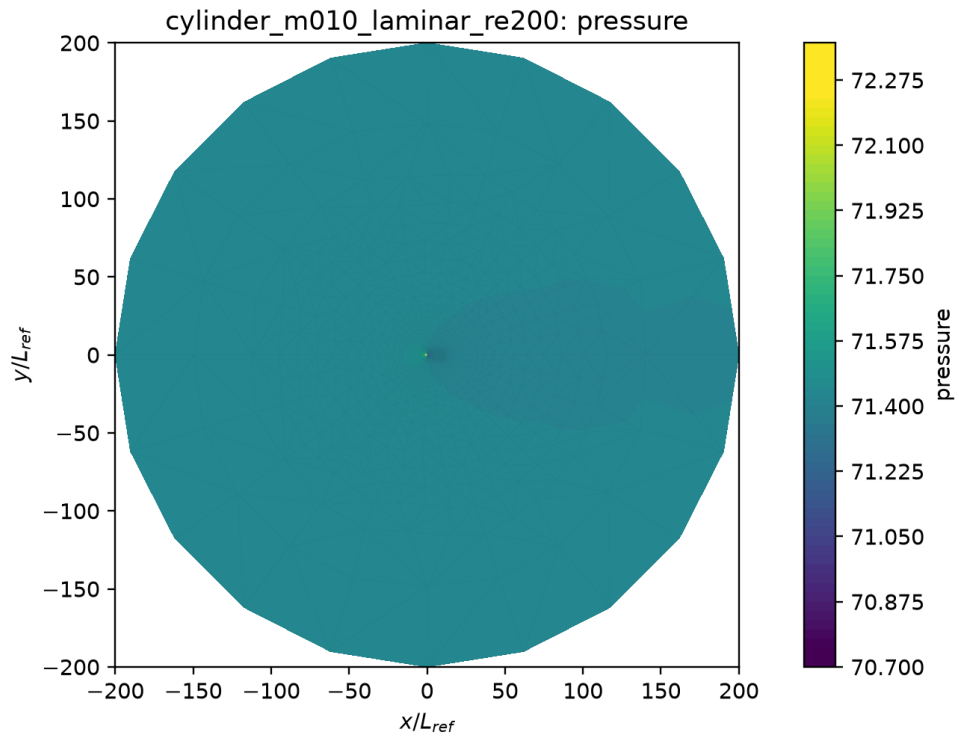


Figure 13: cylinder_m010_laminar_re200: color-filled pressure contour (full domain) derived from field_final.vtk.

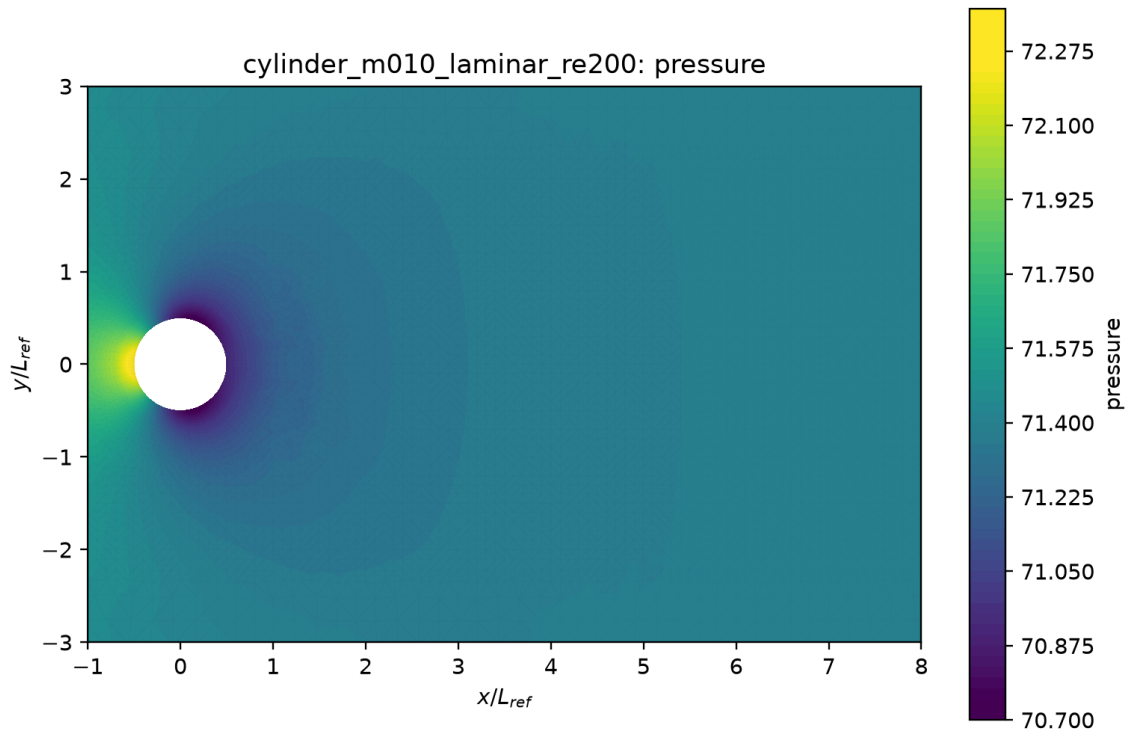


Figure 14: cylinder_m010_laminar_re200: color-filled pressure contour (body/wake zoom) derived from field_final.vtk.

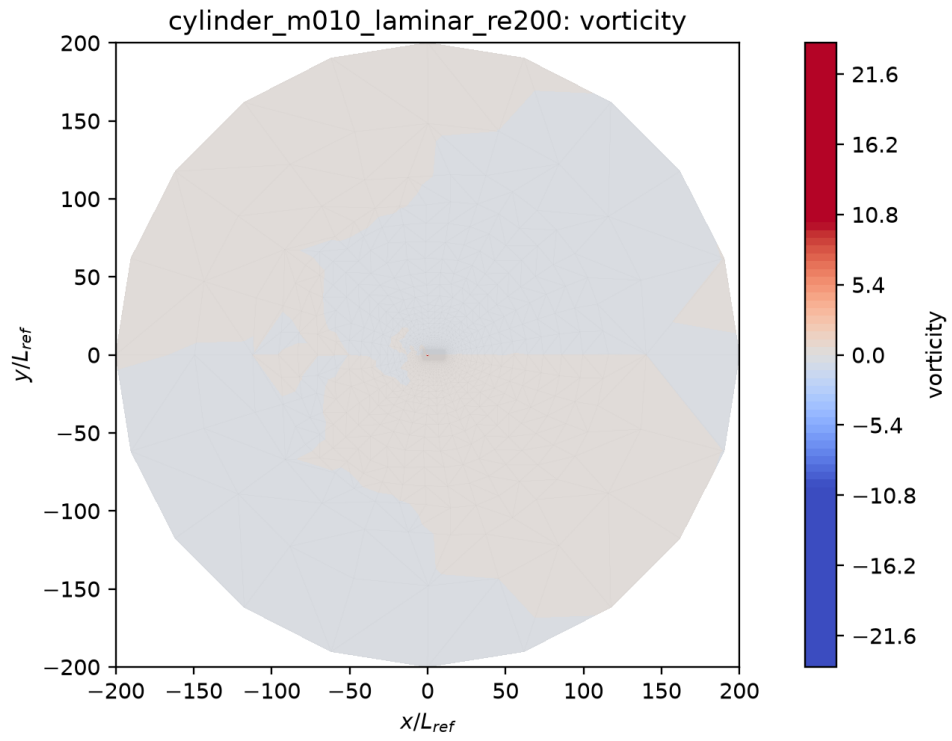


Figure 15: cylinder_m010_laminar_re200: color-filled vorticity contour (full domain) derived from field_final.vtk.

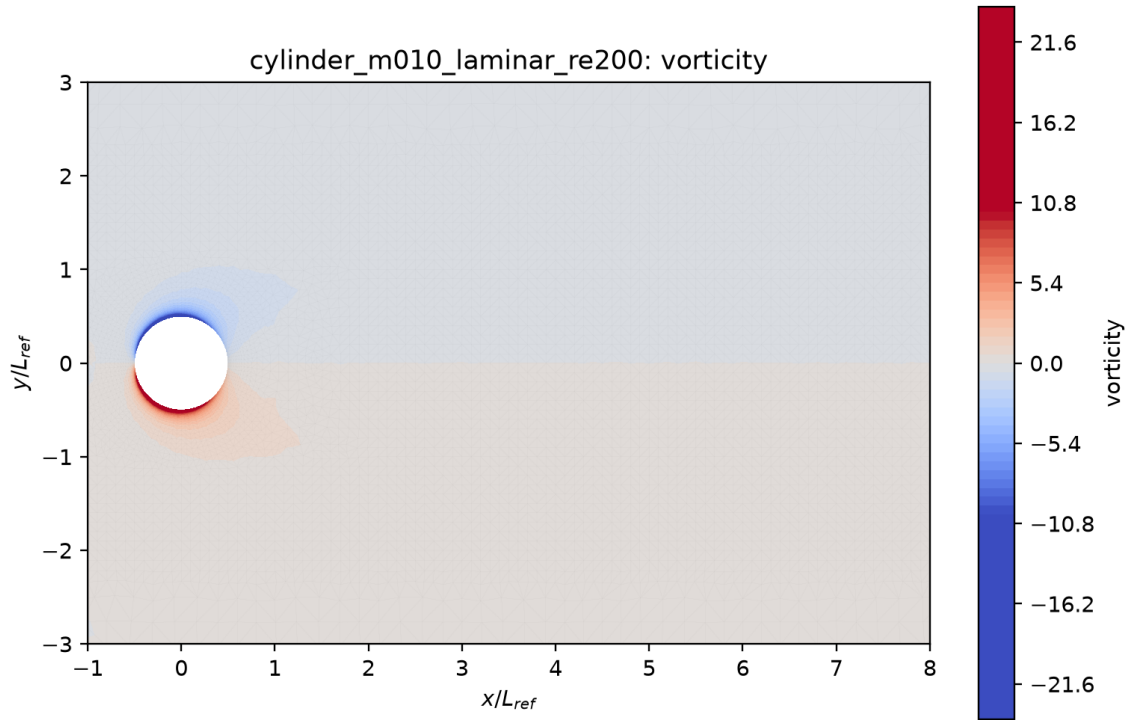


Figure 16: cylinder_m010_laminar_re200: color-filled vorticity contour (body/wake zoom) derived from field_final.vtk.

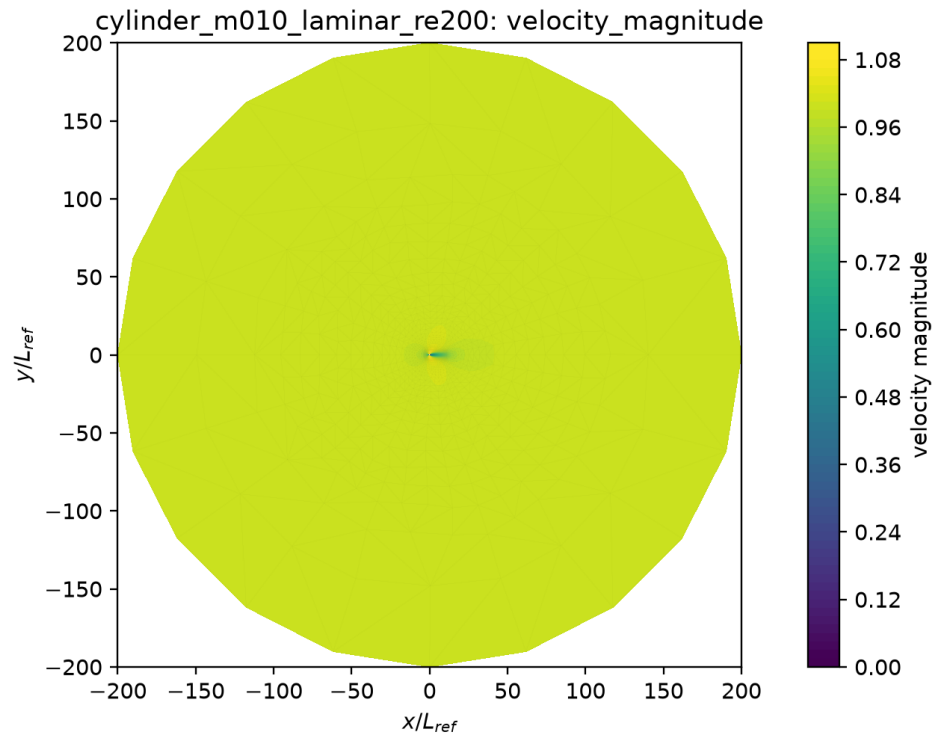


Figure 17: cylinder_m010_laminar_re200: color-filled velocity magnitude contour (full domain) derived from field_final.vtk.

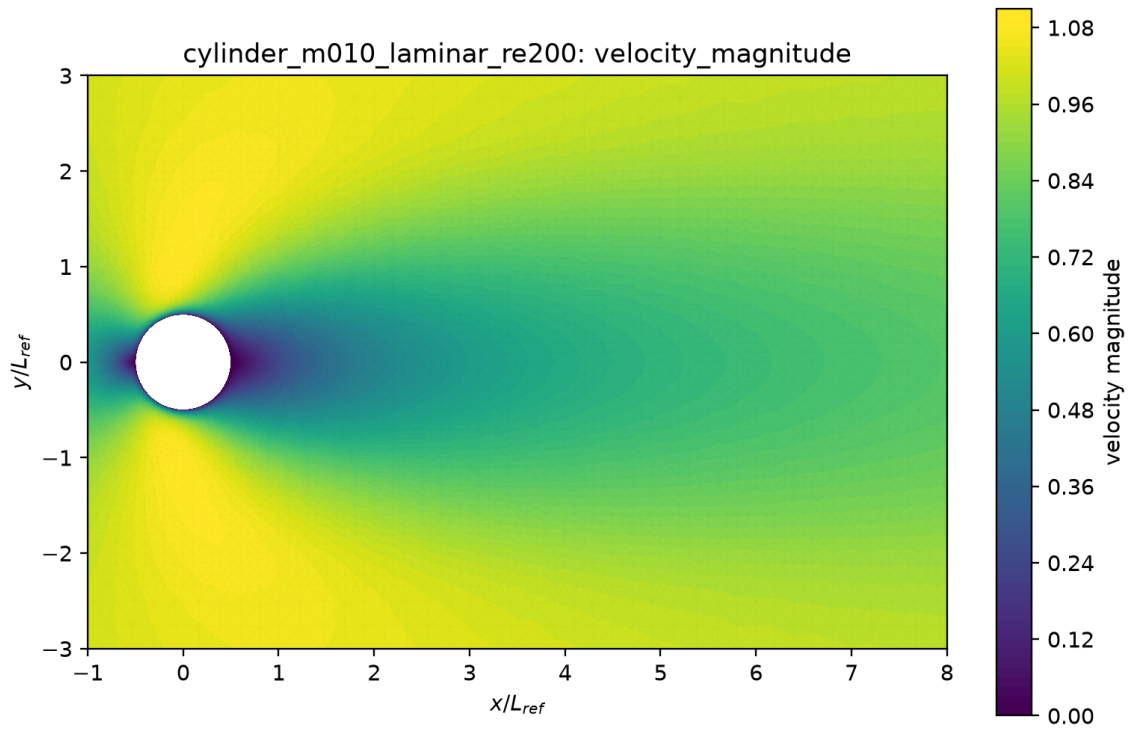


Figure 18: cylinder_m010_laminar_re200: color-filled velocity magnitude contour (body/wake zoom) derived from field_final.vtk.

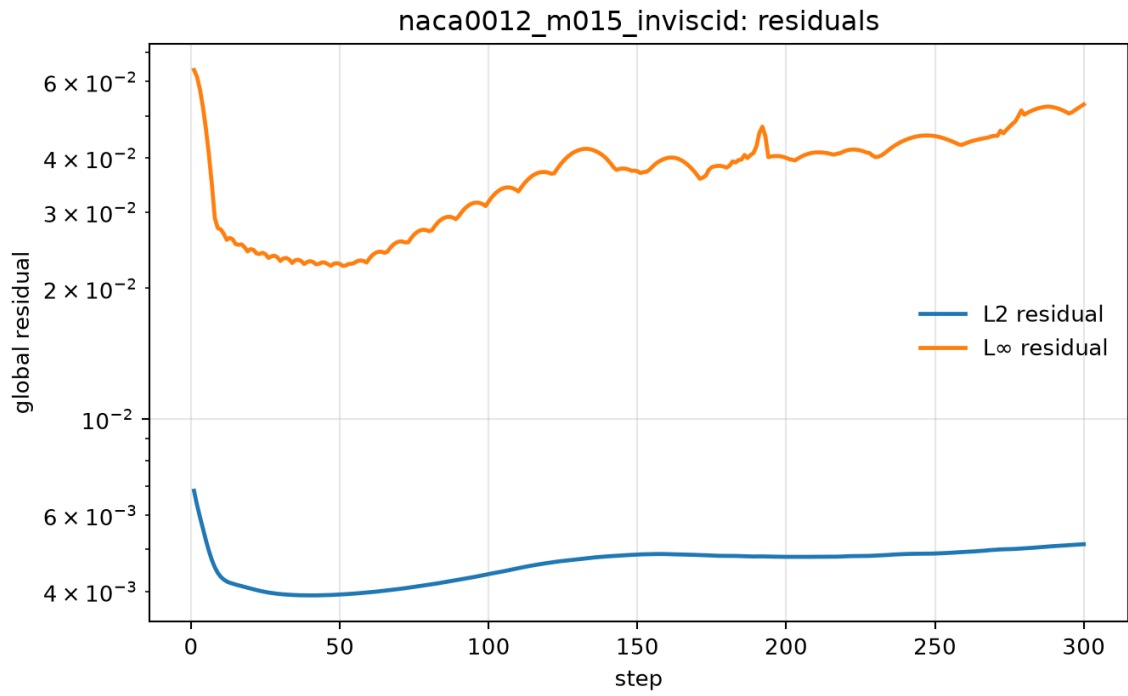


Figure 19: naca0012_m015_inviscid: line plot of L2 and L-infinity global residuals (full domain) derived from residuals.csv.

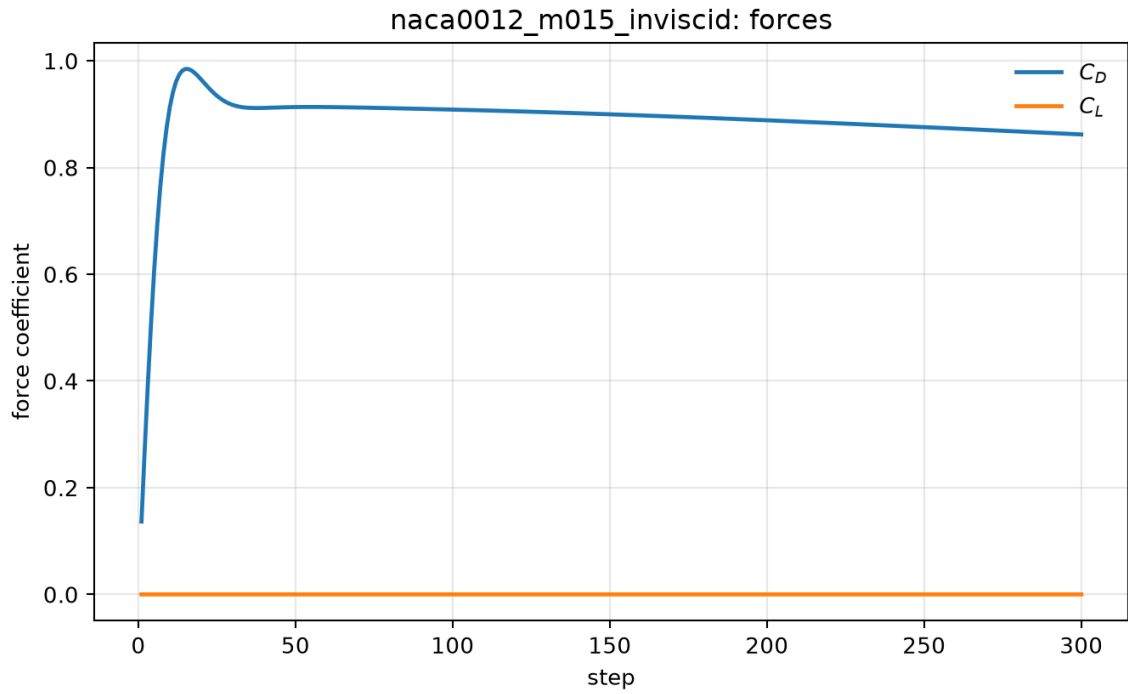


Figure 20: naca0012_m015_inviscid: line plot of drag (C_D) and lift (C_L) coefficients (full domain) derived from forces.csv.

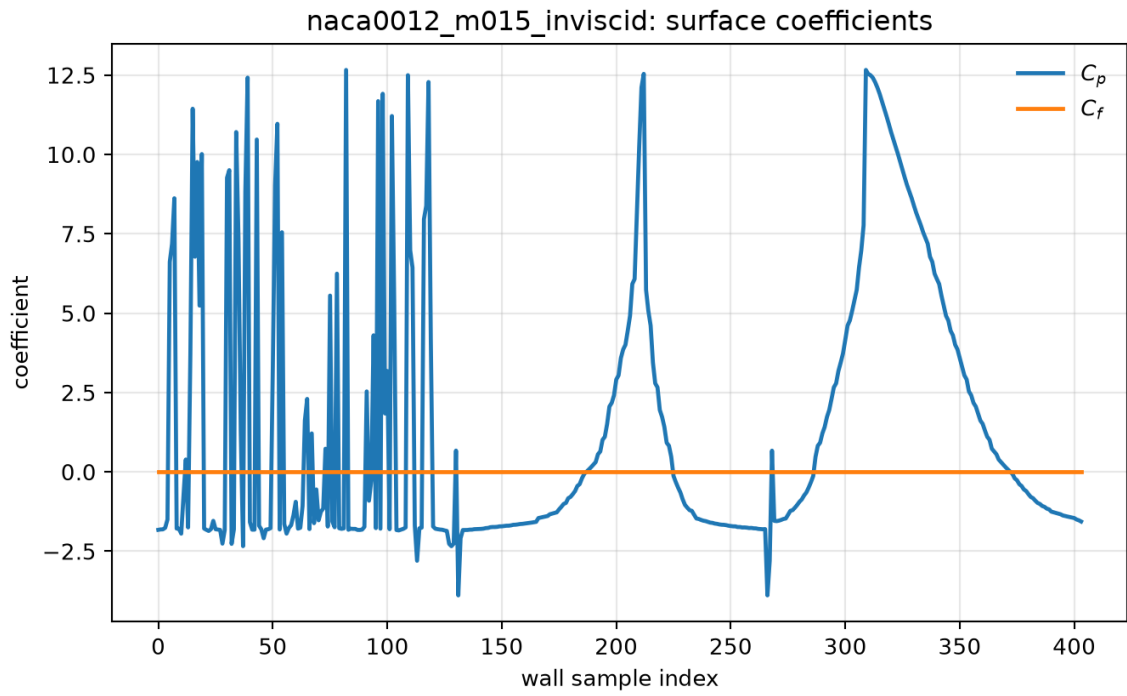


Figure 21: naca0012_m015_inviscid: line plot of pressure (C_p) and skin-friction (C_f) surface coefficients (full domain) derived from surface.csv.

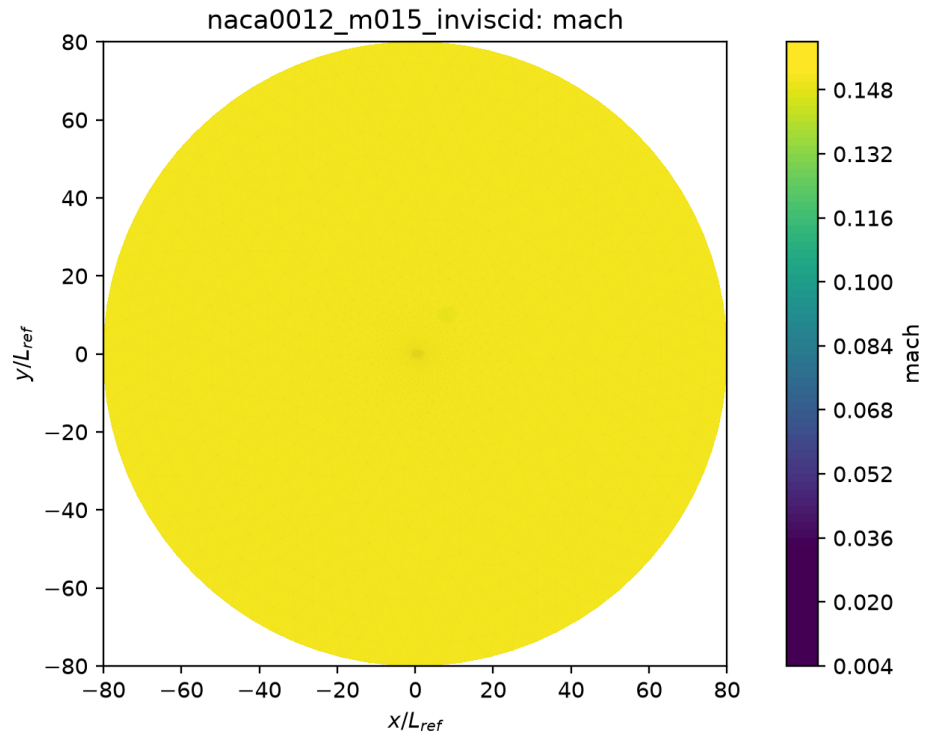


Figure 22: naca0012_m015_inviscid: color-filled mach contour (full domain) derived from field_final.vtk.

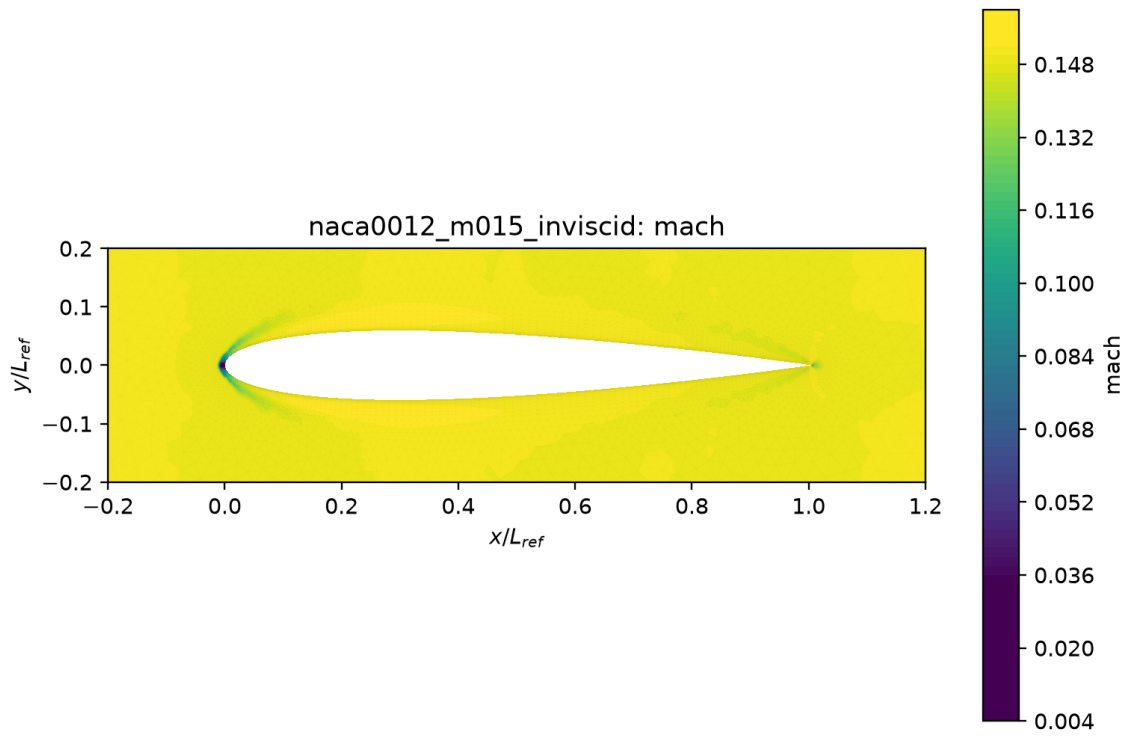


Figure 23: naca0012_m015_inviscid: color-filled mach contour (body/wake zoom) derived from field_final.vtk.

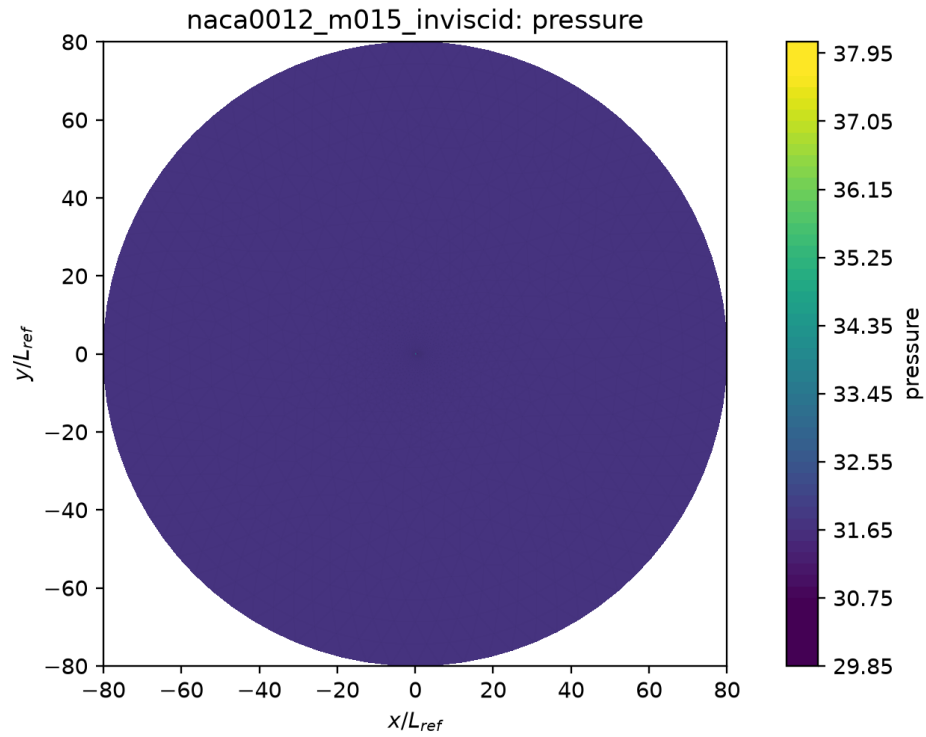


Figure 24: naca0012_m015_inviscid: color-filled pressure contour (full domain) derived from field_final.vtk.

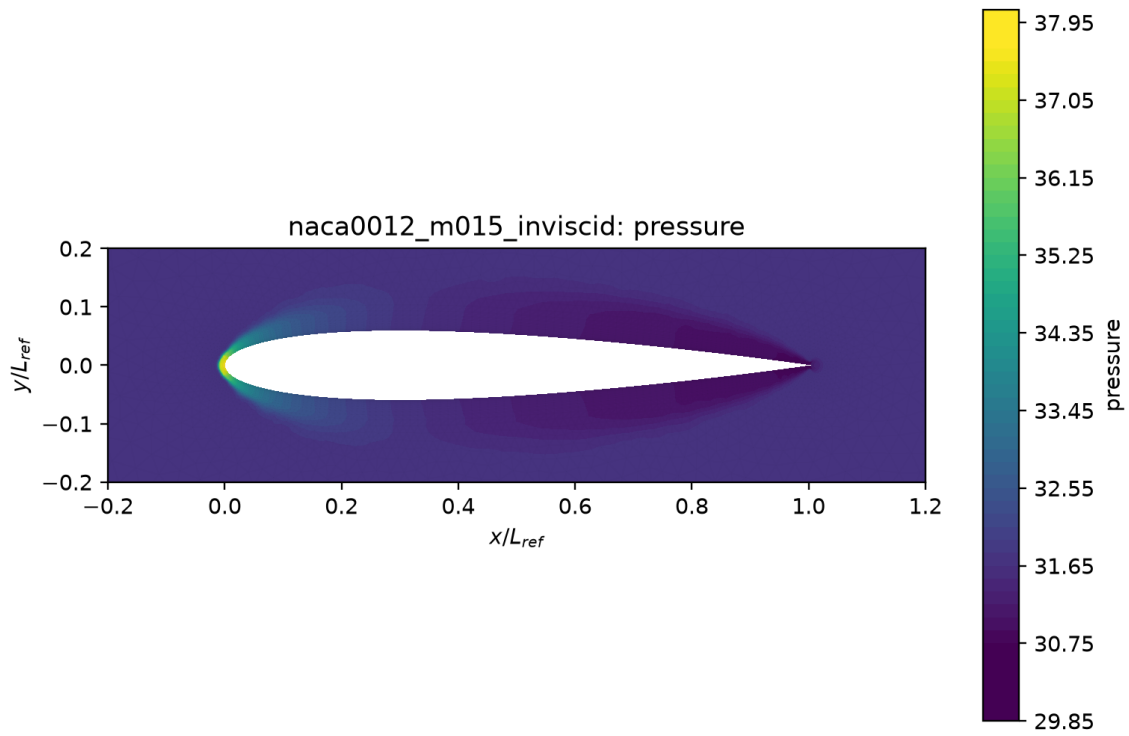


Figure 25: naca0012_m015_inviscid: color-filled pressure contour (body/wake zoom) derived from field_final.vtk.

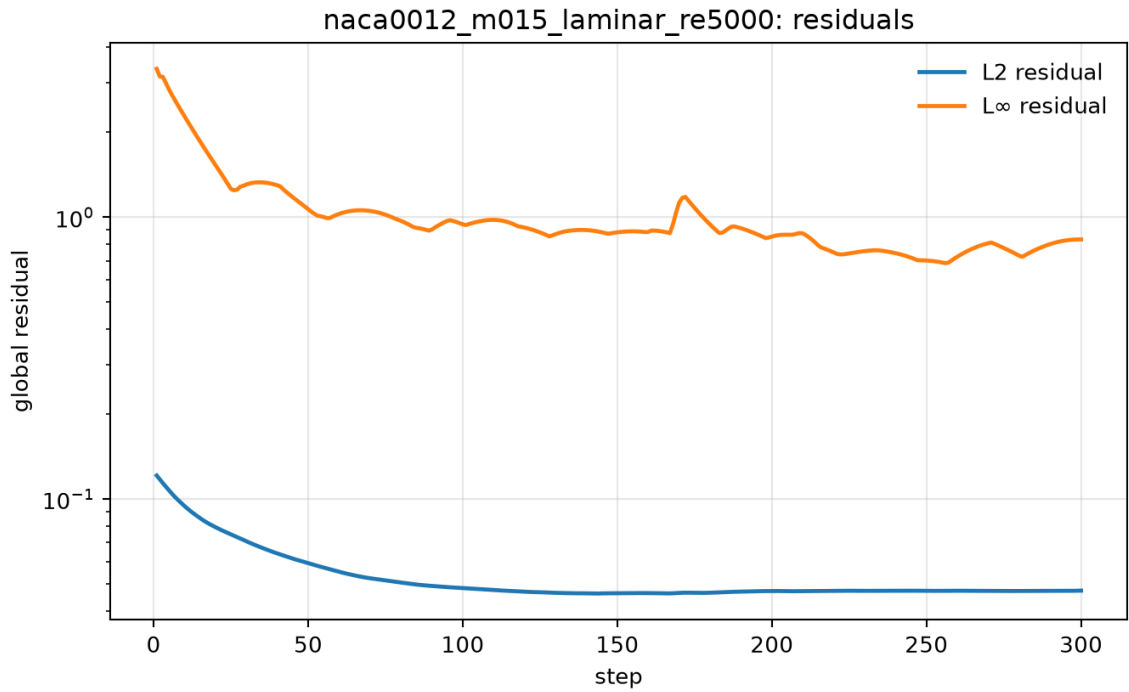


Figure 26: naca0012_m015_laminar_re5000: line plot of L2 and L-infinity global residuals (full domain) derived from residuals.csv.

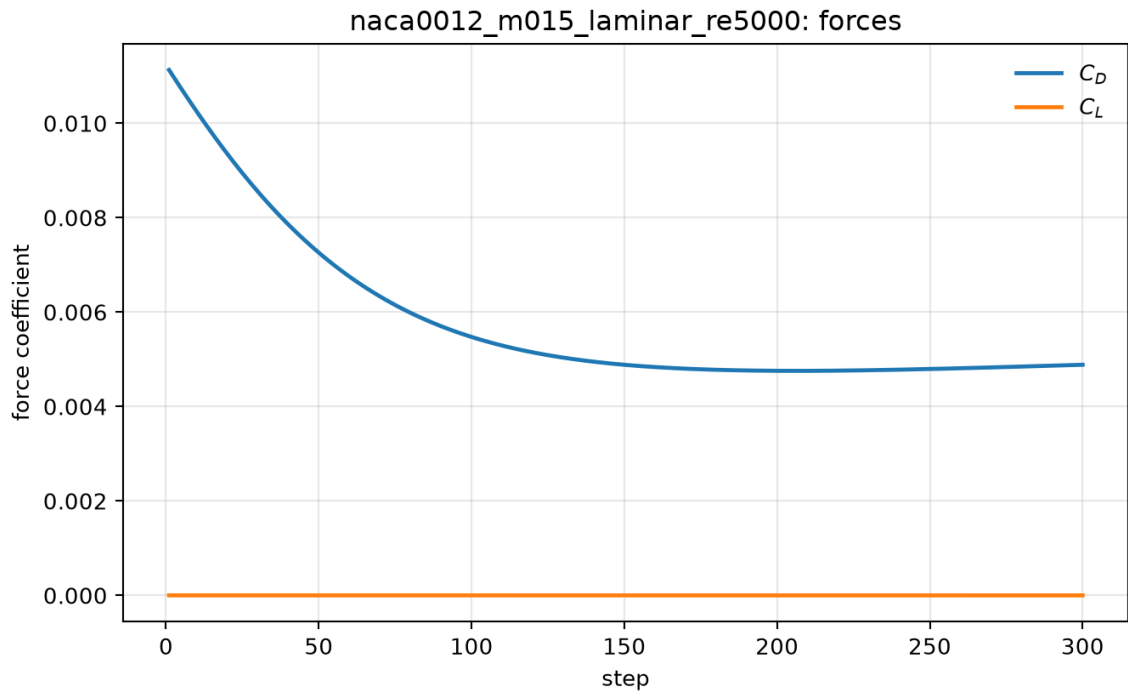


Figure 27: naca0012_m015_laminar_re5000: line plot of drag (C_D) and lift (C_L) coefficients (full domain) derived from forces.csv.

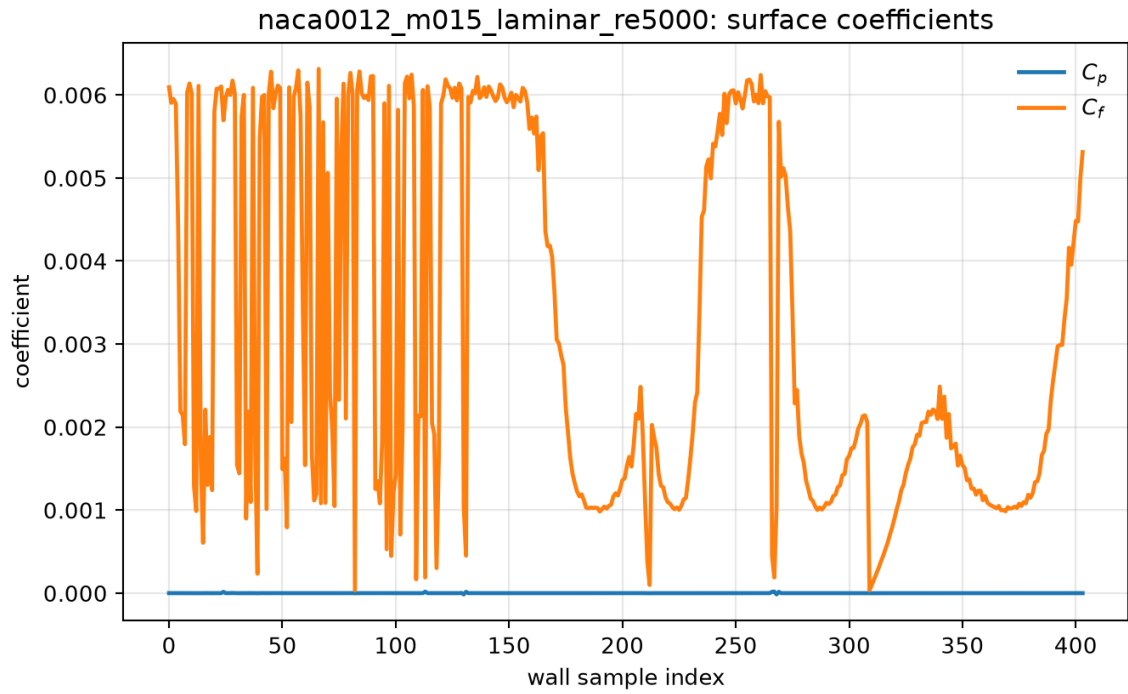


Figure 28: naca0012_m015_laminar_re5000: line plot of pressure (C_p) and skin-friction (C_f) surface coefficients (full domain) derived from surface.csv.

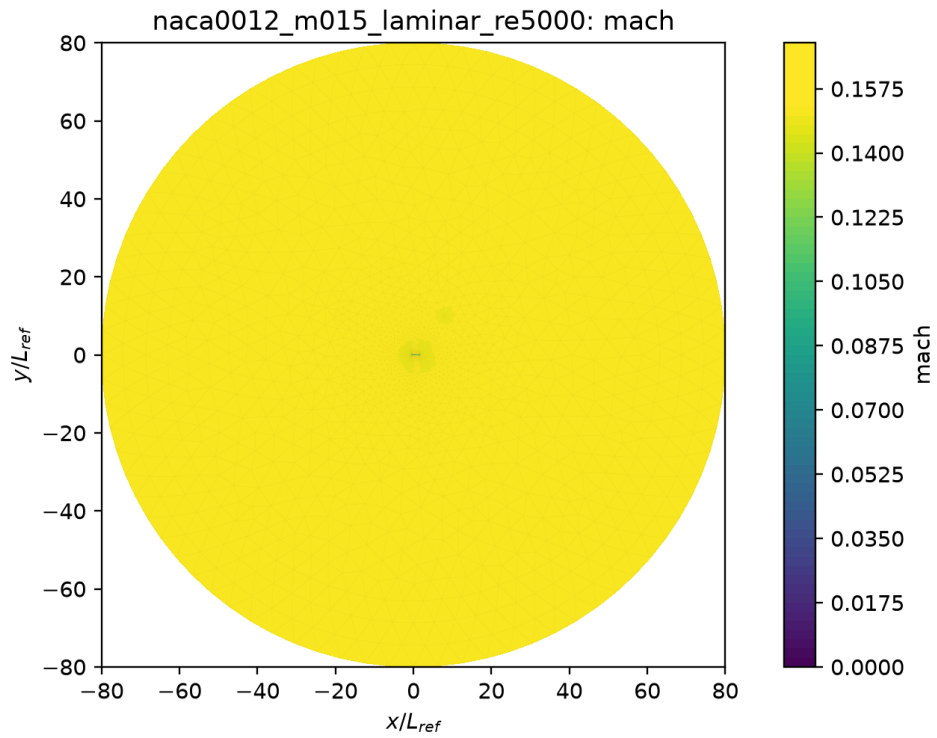


Figure 29: naca0012_m015_laminar_re5000: color-filled mach contour (full domain) derived from field_final.vtk.

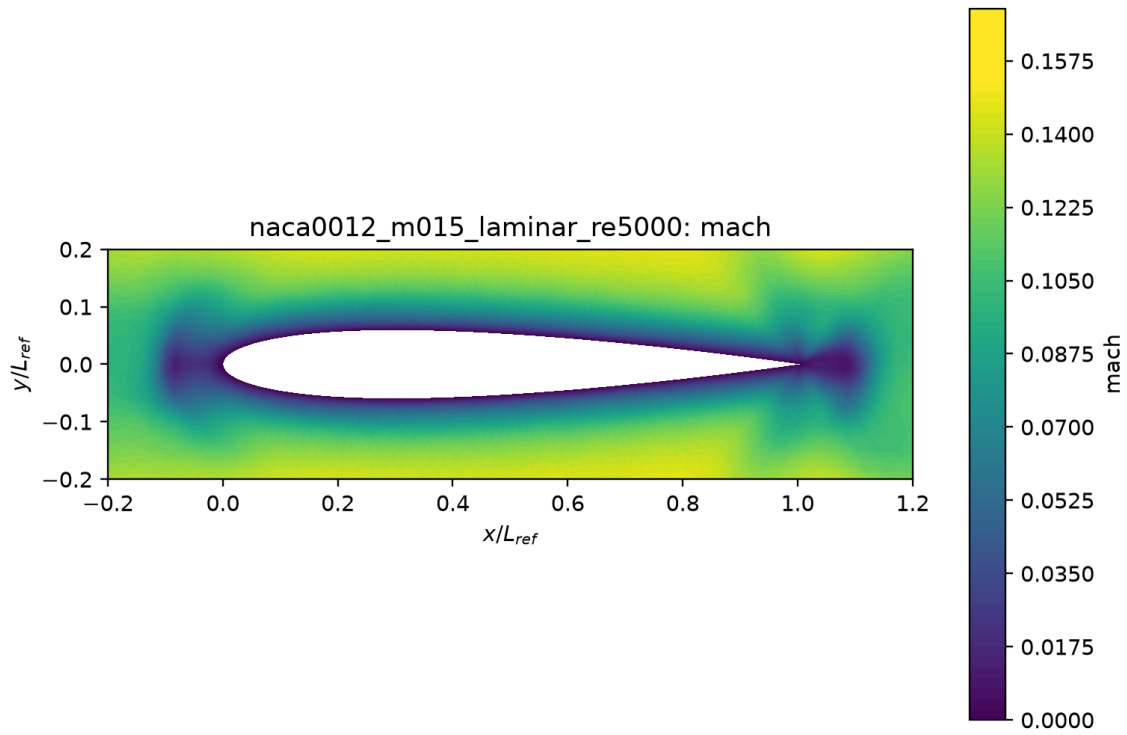


Figure 30: naca0012_m015_laminar_re5000: color-filled mach contour (body/wake zoom) derived from field_final.vtk.

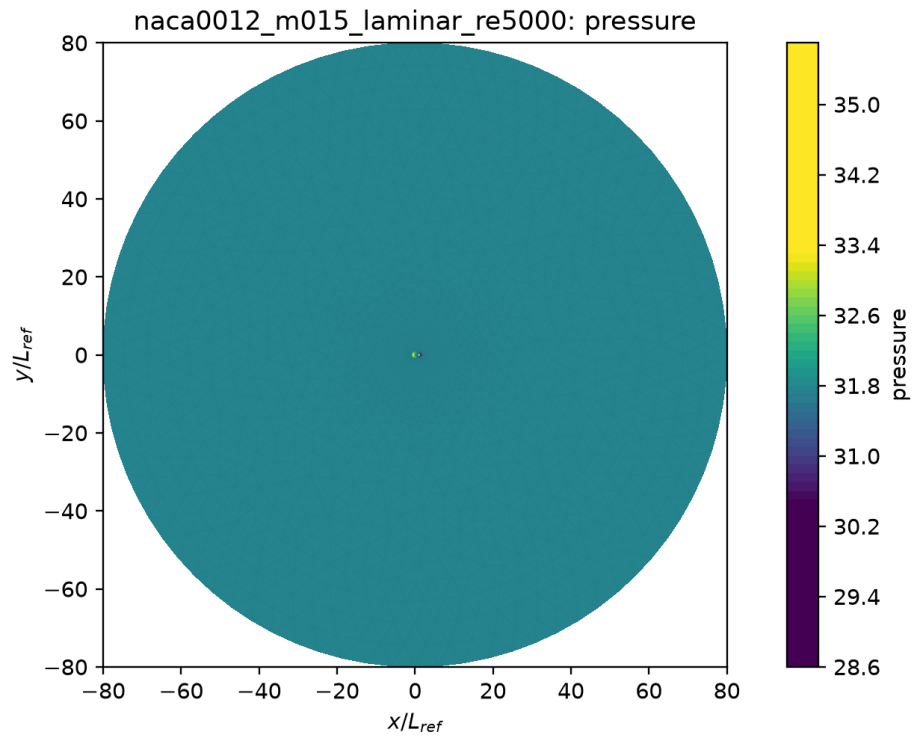


Figure 31: naca0012_m015_laminar_re5000: color-filled pressure contour (full domain) derived from field_final.vtk.

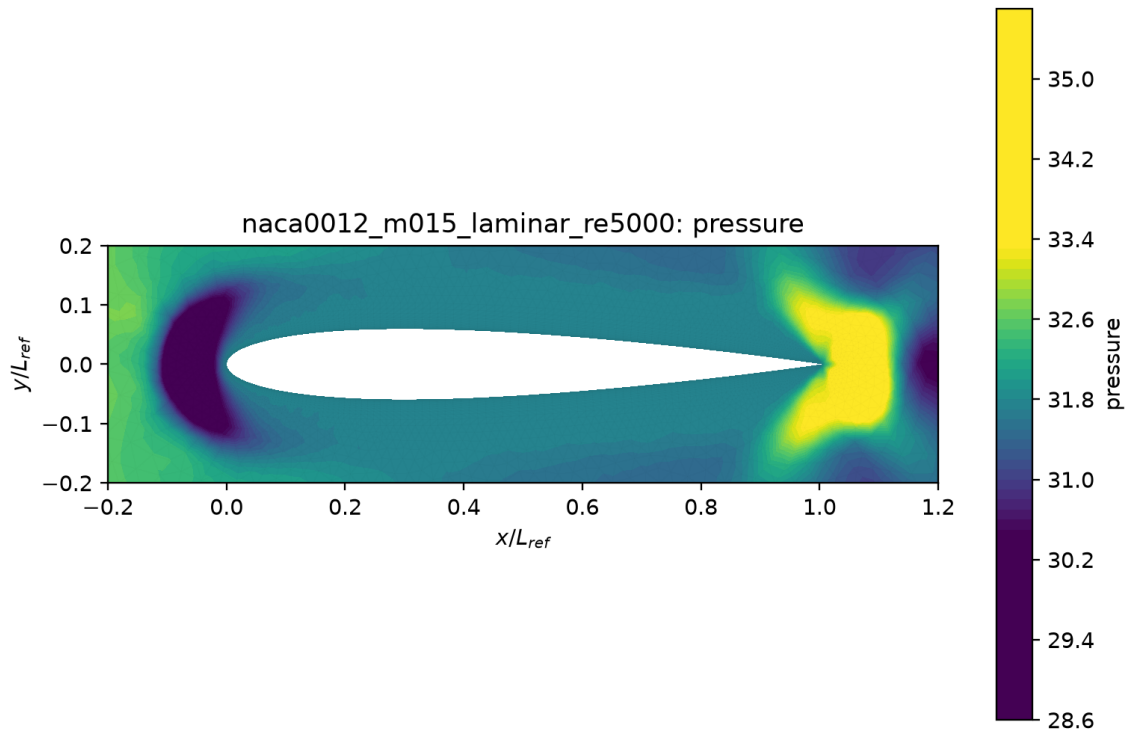


Figure 32: naca0012_m015_laminar_re5000: color-filled pressure contour (body/wake zoom) derived from field_final.vtk.

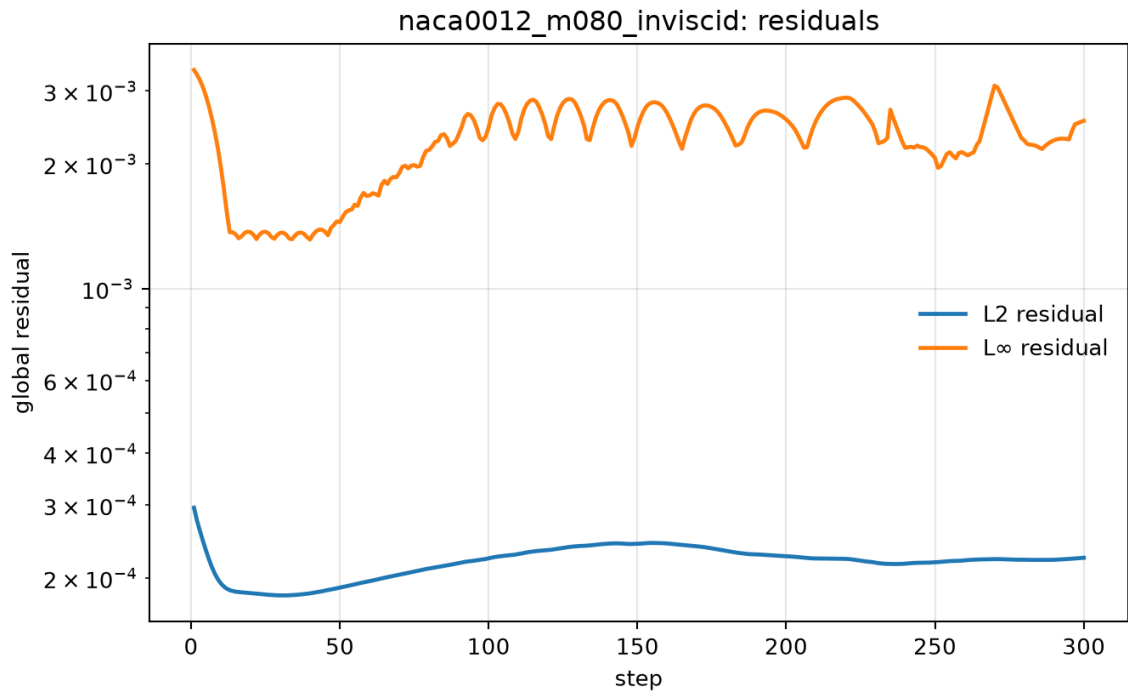


Figure 33: naca0012_m080_inviscid: line plot of L2 and L-infinity global residuals (full domain) derived from residuals.csv.

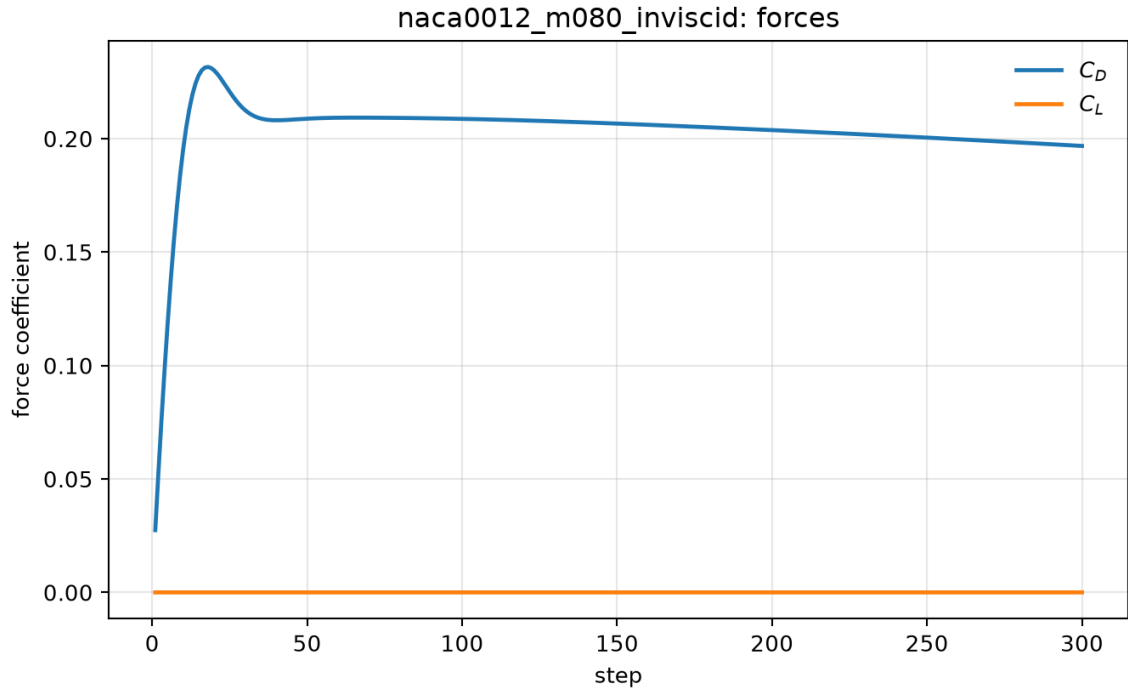


Figure 34: naca0012_m080_inviscid: line plot of drag (C_D) and lift (C_L) coefficients (full domain) derived from forces.csv.

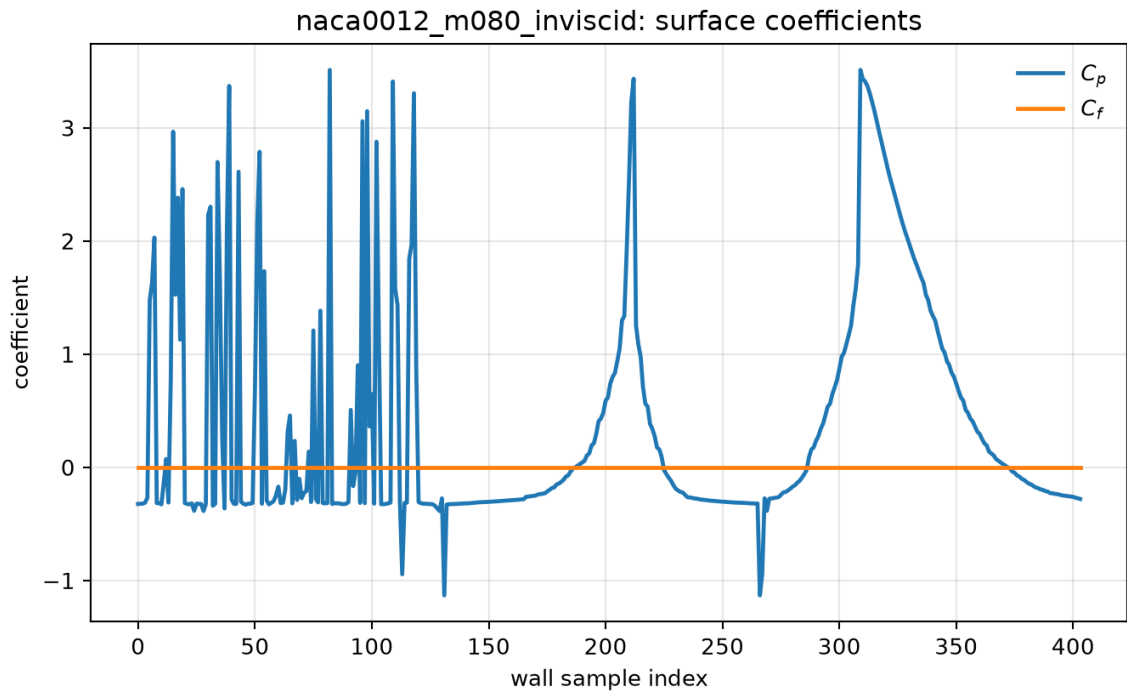


Figure 35: naca0012_m080_inviscid: line plot of pressure (C_p) and skin-friction (C_f) surface coefficients (full domain) derived from surface.csv.

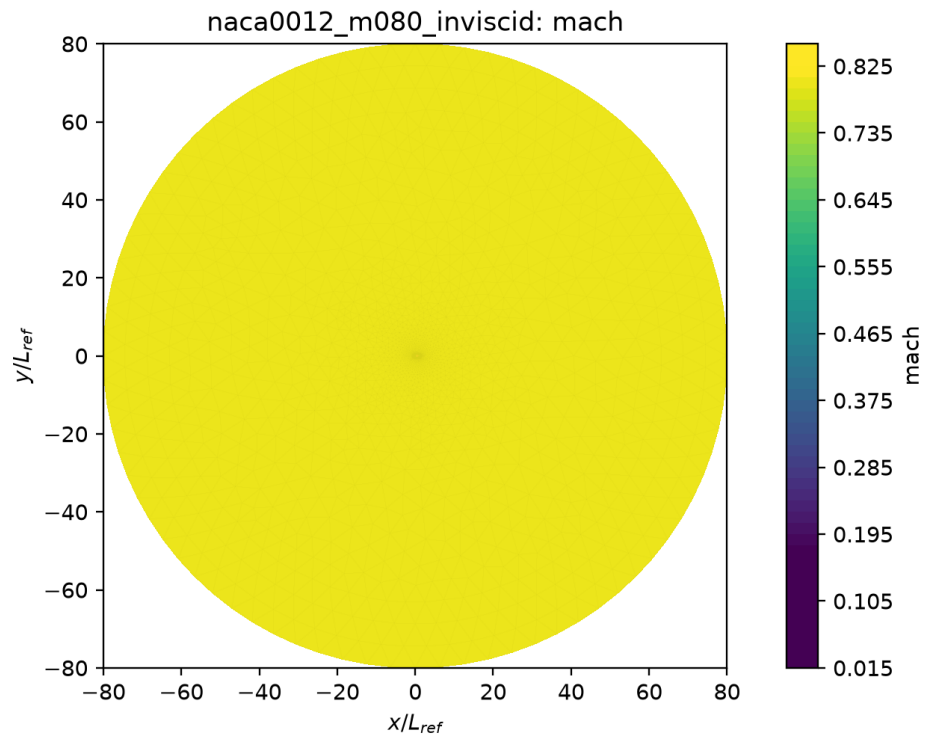


Figure 36: naca0012_m080_inviscid: color-filled mach contour (full domain) derived from field_final.vtk.

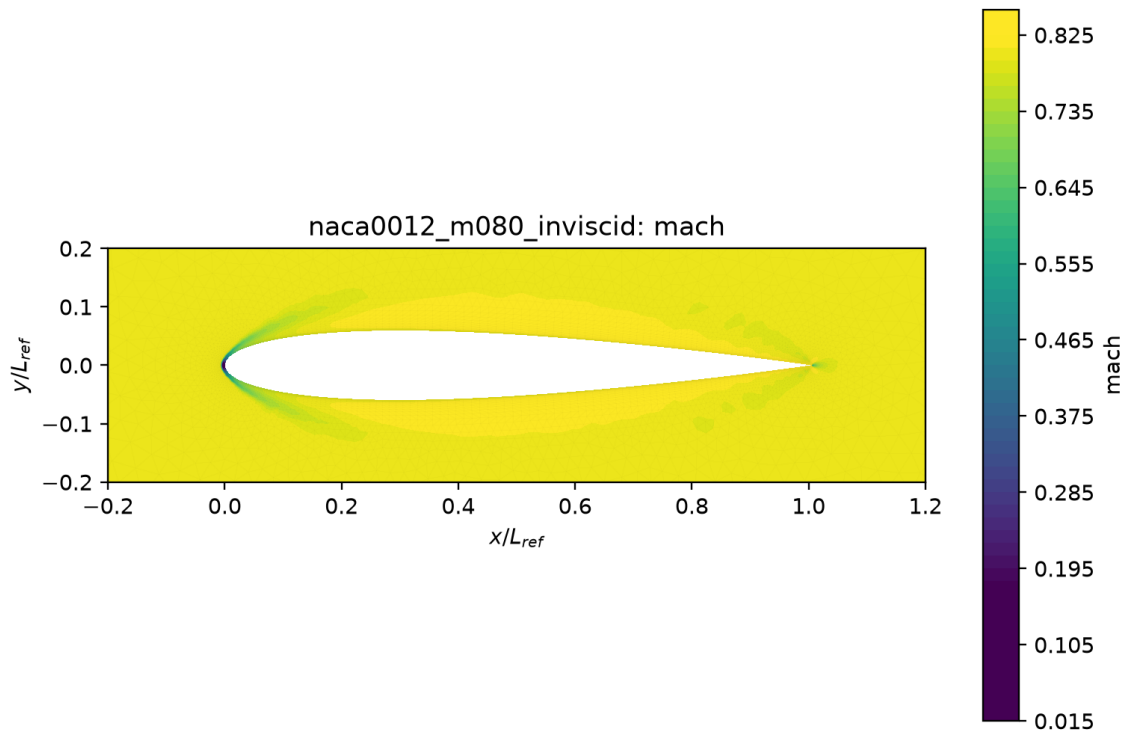


Figure 37: naca0012_m080_inviscid: color-filled mach contour (body/wake zoom) derived from field_final.vtk.

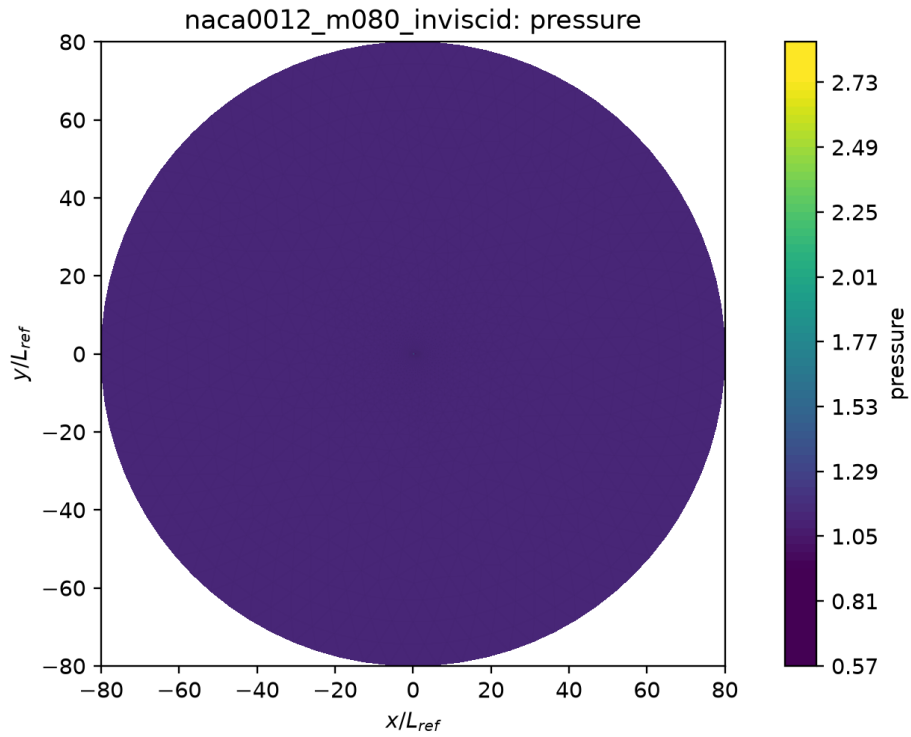


Figure 38: naca0012_m080_inviscid: color-filled pressure contour (full domain) derived from field_final.vtk.

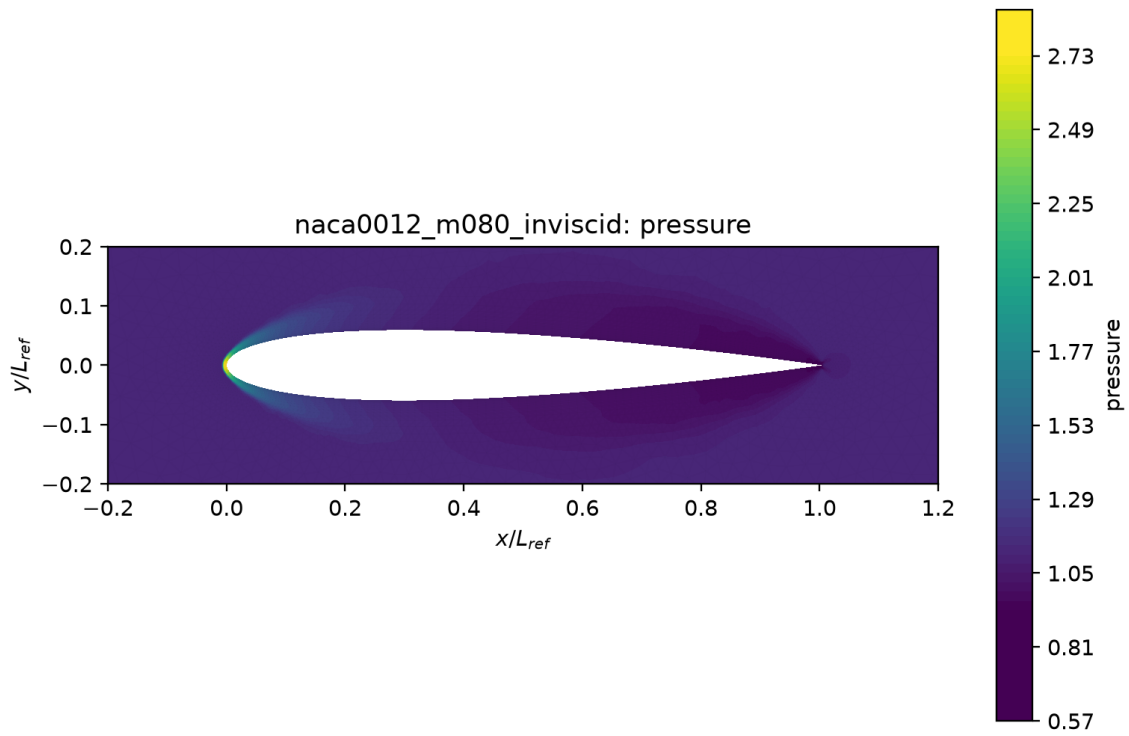


Figure 39: naca0012_m080_inviscid: color-filled pressure contour (body/wake zoom) derived from field_final.vtk.

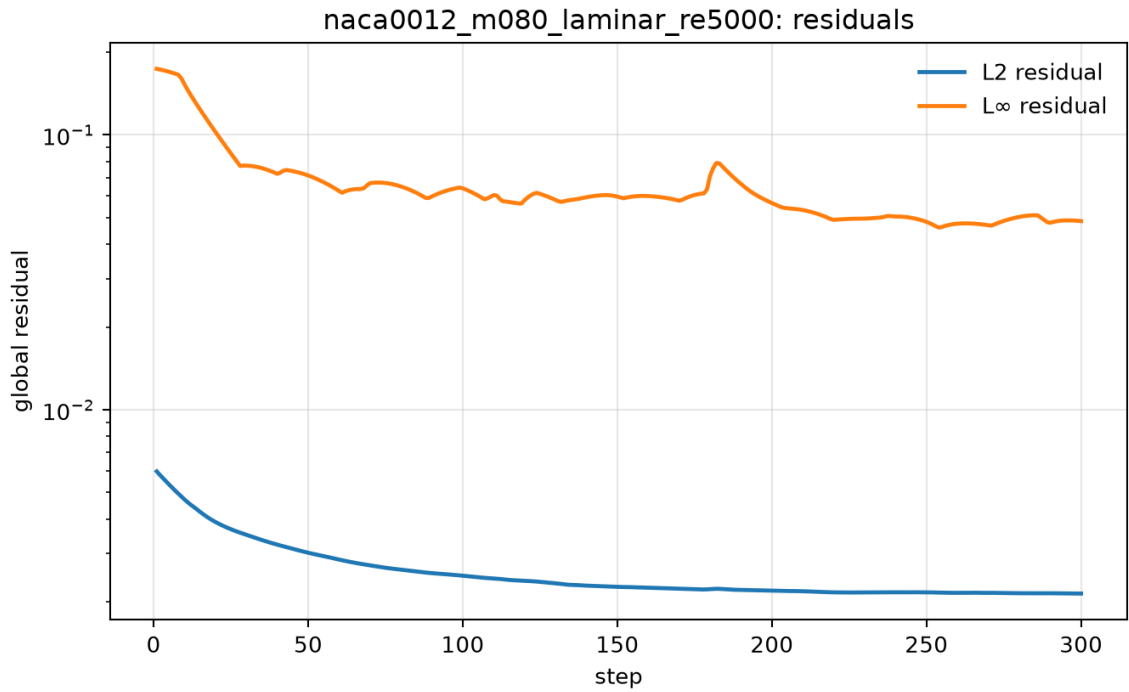


Figure 40: naca0012_m080_laminar_re5000: line plot of L2 and L-infinity global residuals (full domain) derived from residuals.csv.

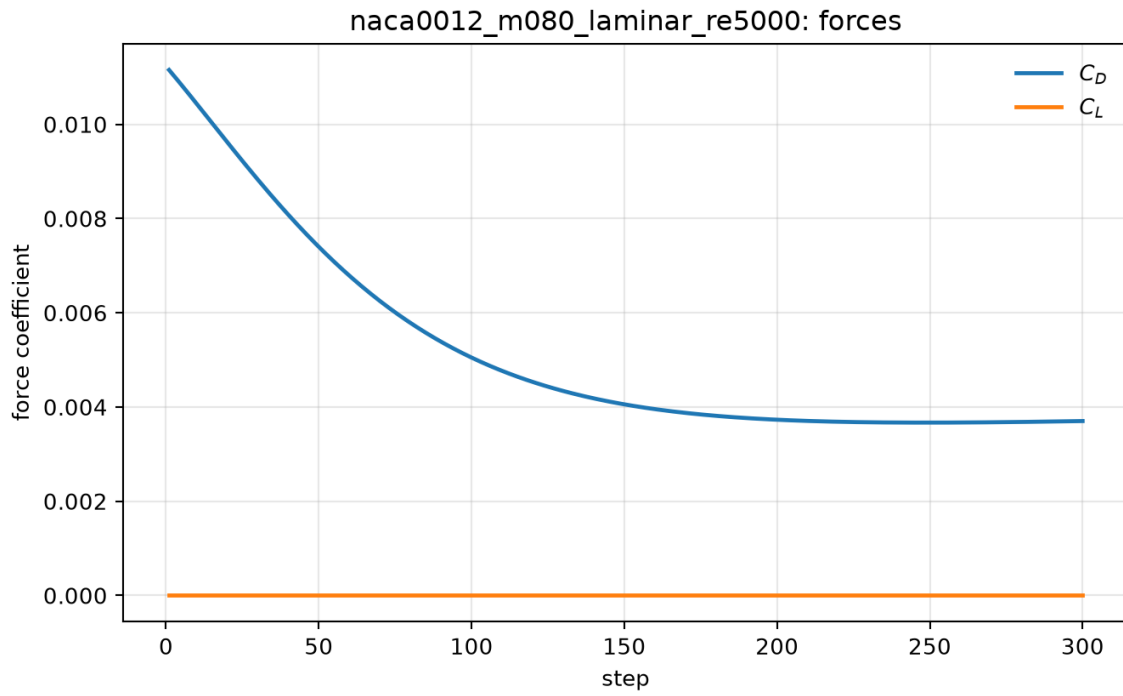


Figure 41: naca0012_m080_laminar_re5000: line plot of drag (C_D) and lift (C_L) coefficients (full domain) derived from forces.csv.

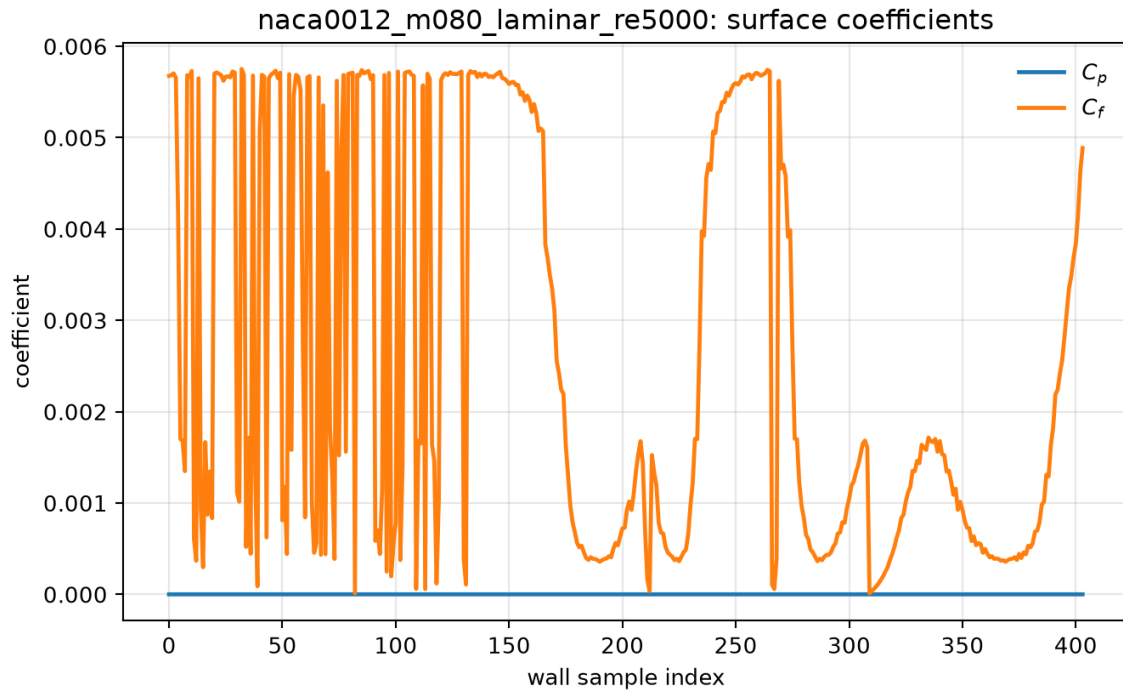


Figure 42: naca0012_m080_laminar_re5000: line plot of pressure (C_p) and skin-friction (C_f) surface coefficients (full domain) derived from surface.csv.

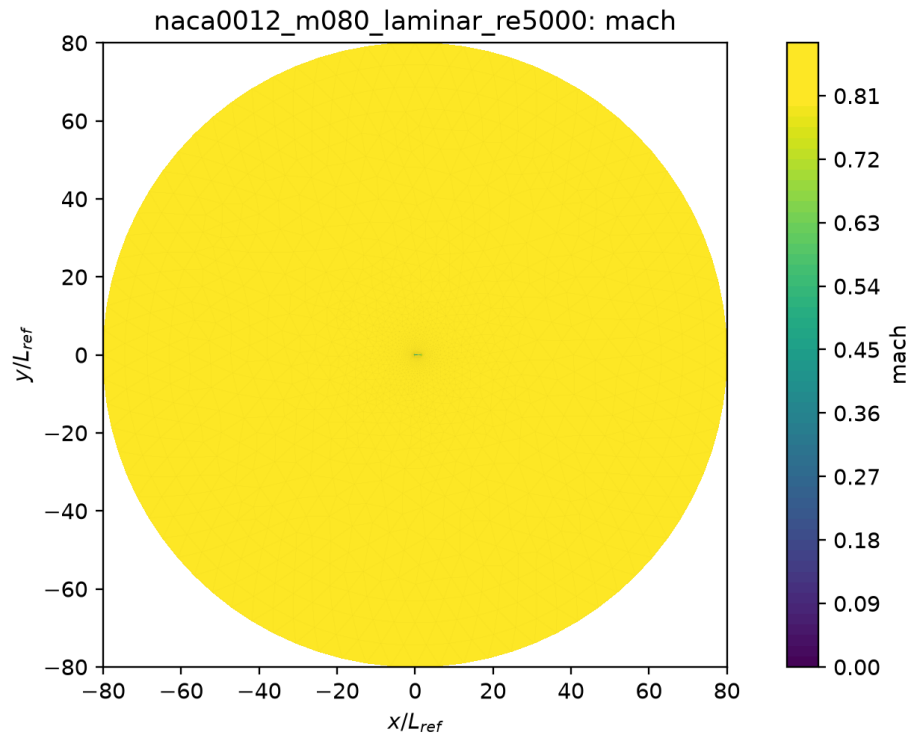


Figure 43: naca0012_m080_laminar_re5000: color-filled mach contour (full domain) derived from field_final.vtk.

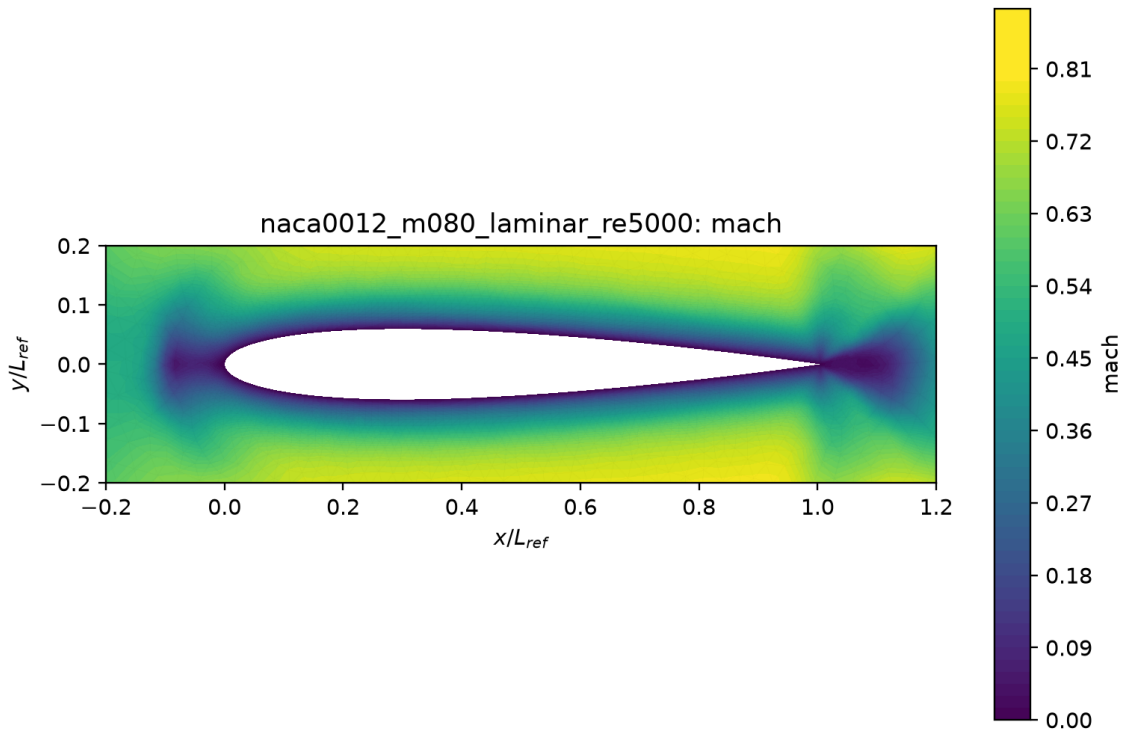


Figure 44: naca0012_m080_laminar_re5000: color-filled mach contour (body/wake zoom) derived from field_final.vtk.

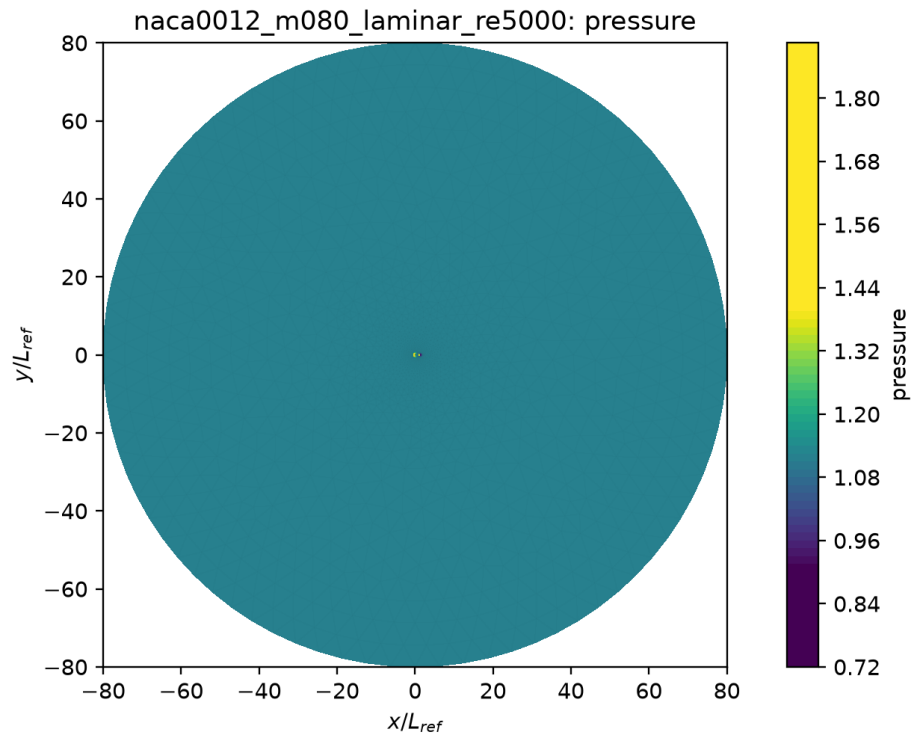


Figure 45: naca0012_m080_laminar_re5000: color-filled pressure contour (full domain) derived from field_final.vtk.

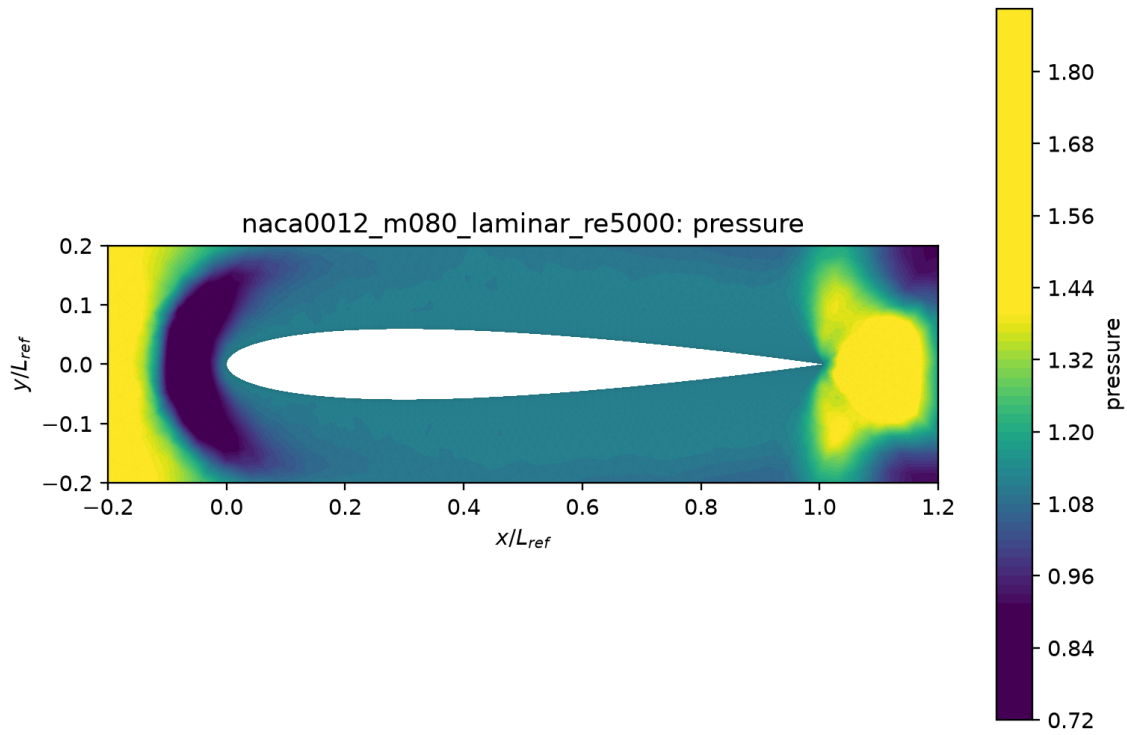


Figure 46: naca0012_m080_laminar_re5000: color-filled pressure contour (body/wake zoom) derived from field_final.vtk.

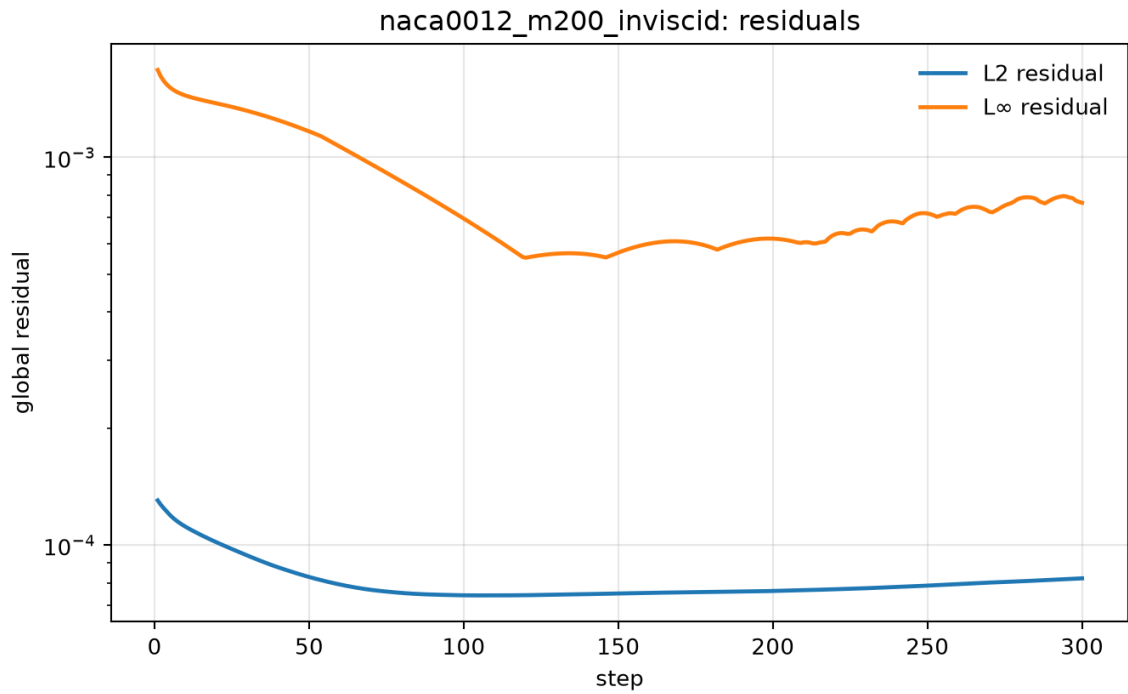


Figure 47: naca0012_m200_inviscid: line plot of L2 and L-infinity global residuals (full domain) derived from residuals.csv.

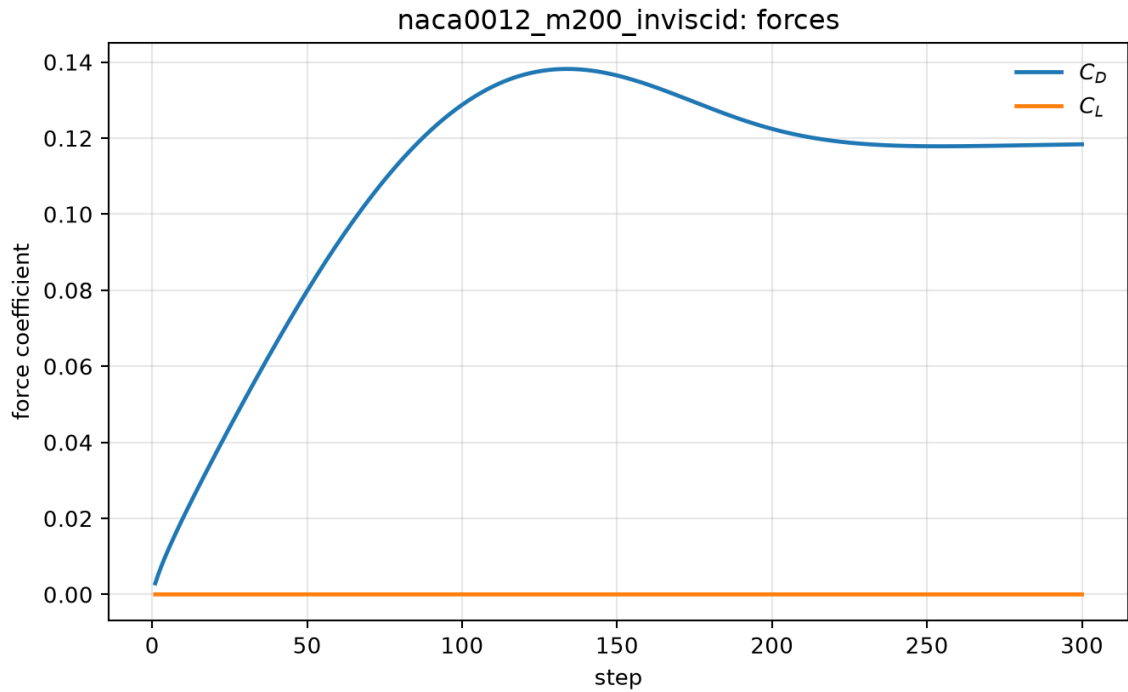


Figure 48: naca0012_m200_inviscid: line plot of drag (C_D) and lift (C_L) coefficients (full domain) derived from forces.csv.

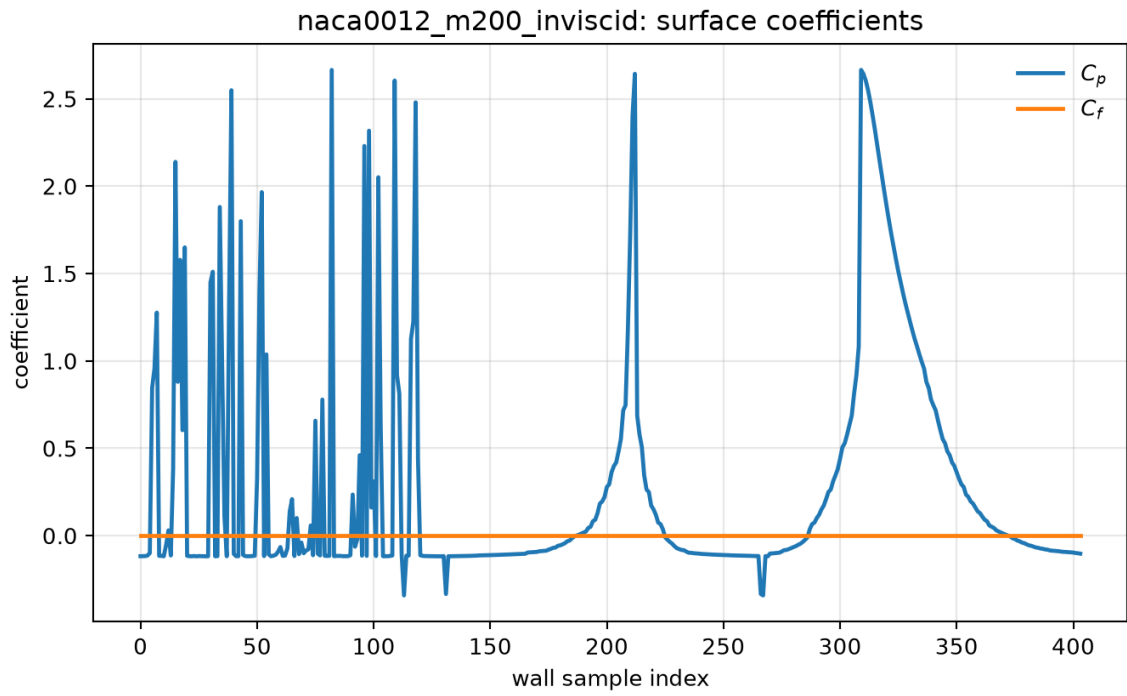


Figure 49: naca0012_m200_inviscid: line plot of pressure (C_p) and skin-friction (C_f) surface coefficients (full domain) derived from surface.csv.

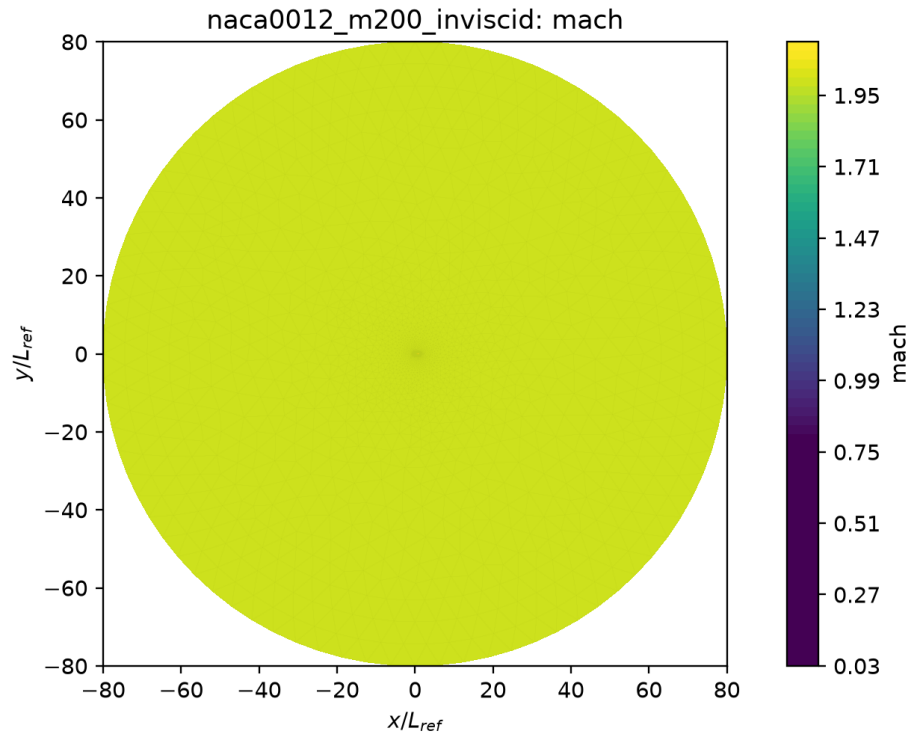


Figure 50: naca0012_m200_inviscid: color-filled mach contour (full domain) derived from field_final.vtk.

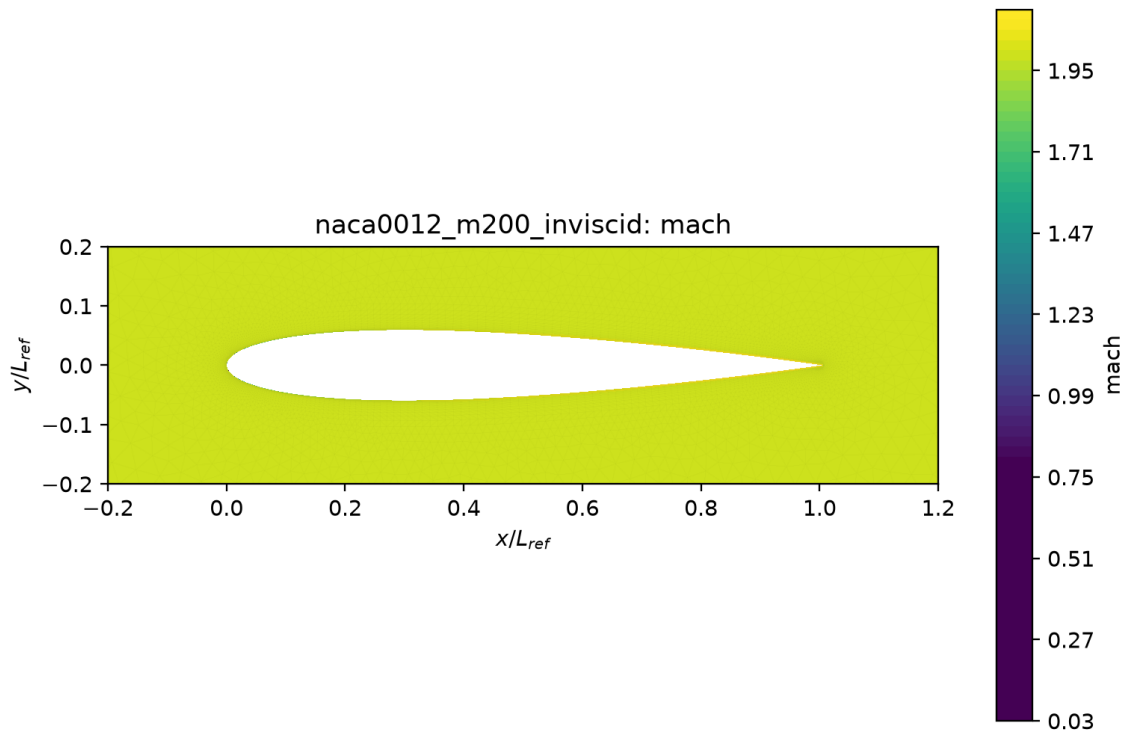


Figure 51: naca0012_m200_inviscid: color-filled mach contour (body/wake zoom) derived from field_final.vtk.

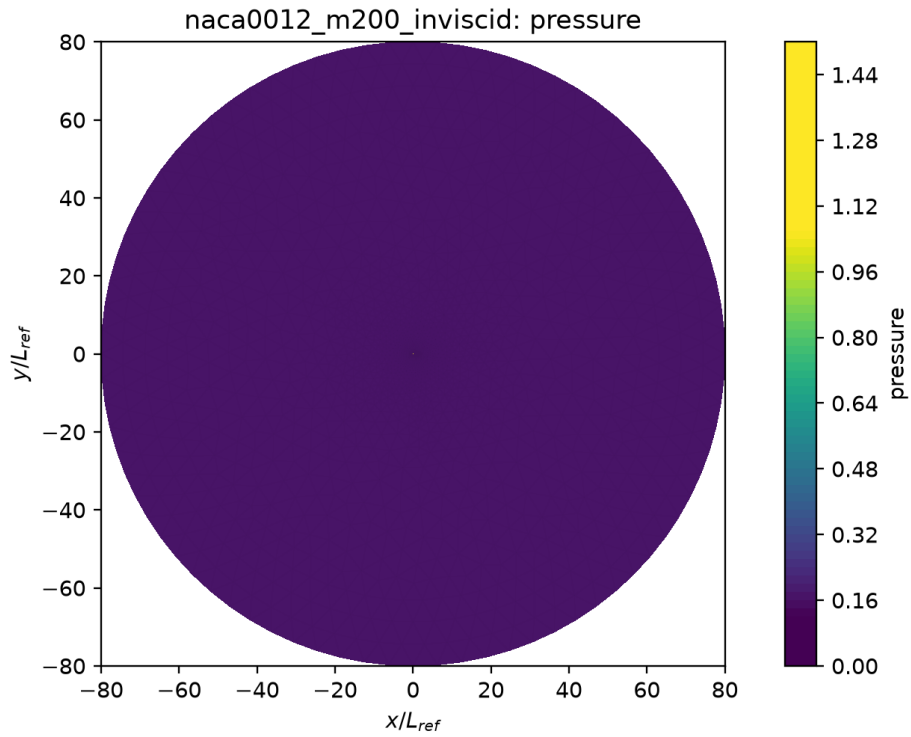


Figure 52: naca0012_m200_inviscid: color-filled pressure contour (full domain) derived from field_final.vtk.

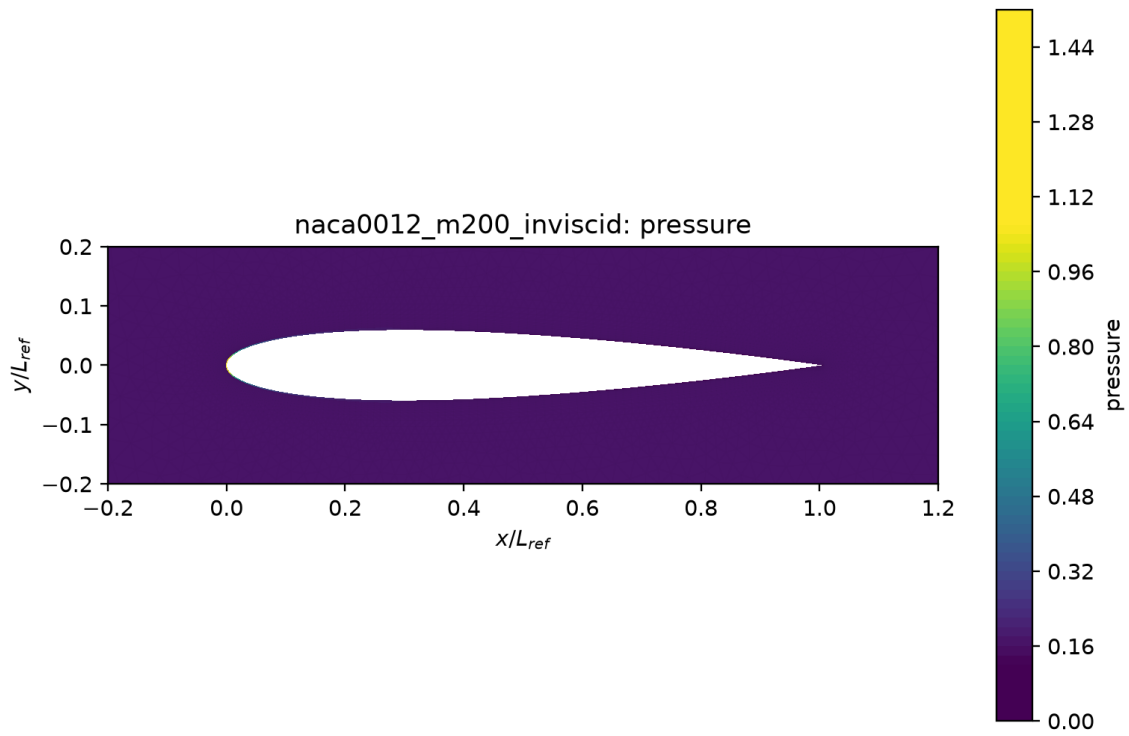


Figure 53: naca0012_m200_inviscid: color-filled pressure contour (body/wake zoom) derived from field_final.vtk.

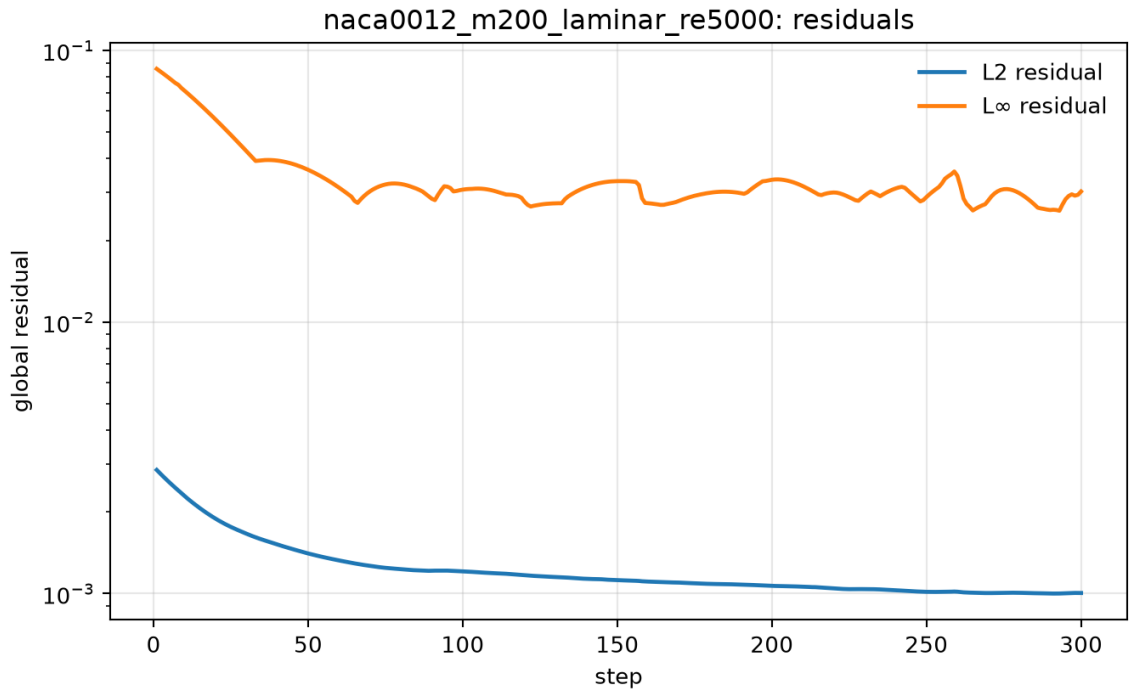


Figure 54: naca0012_m200_laminar_re5000: line plot of L2 and L-infinity global residuals (full domain) derived from residuals.csv.

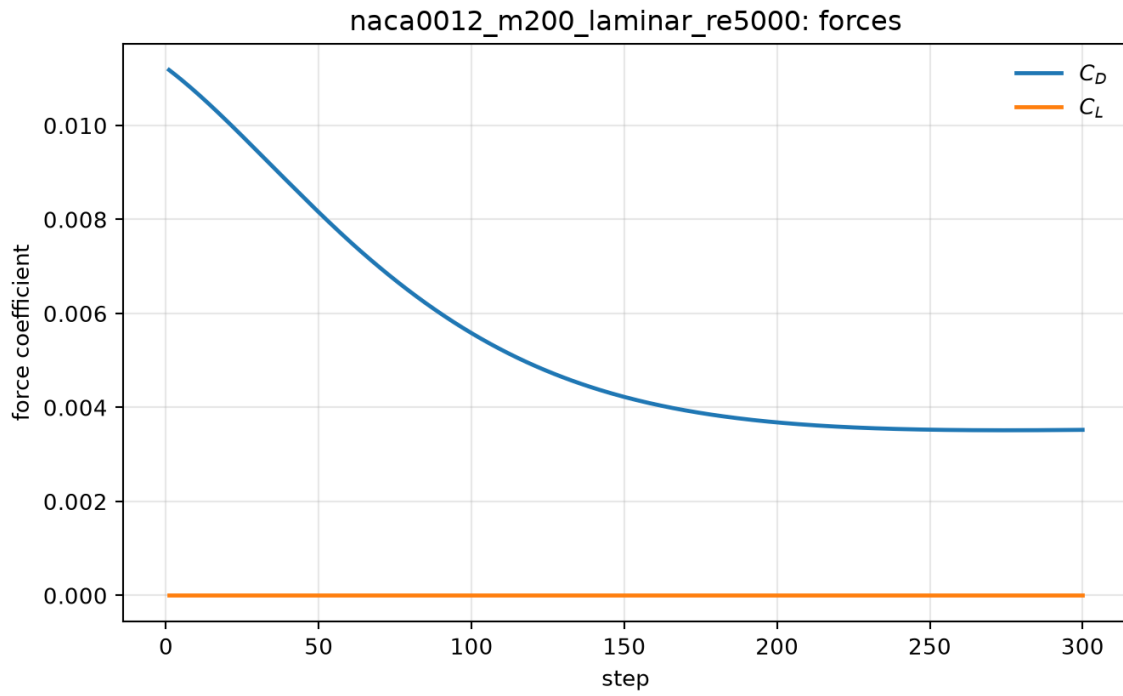


Figure 55: naca0012_m200_laminar_re5000: line plot of drag (C_D) and lift (C_L) coefficients (full domain) derived from forces.csv.

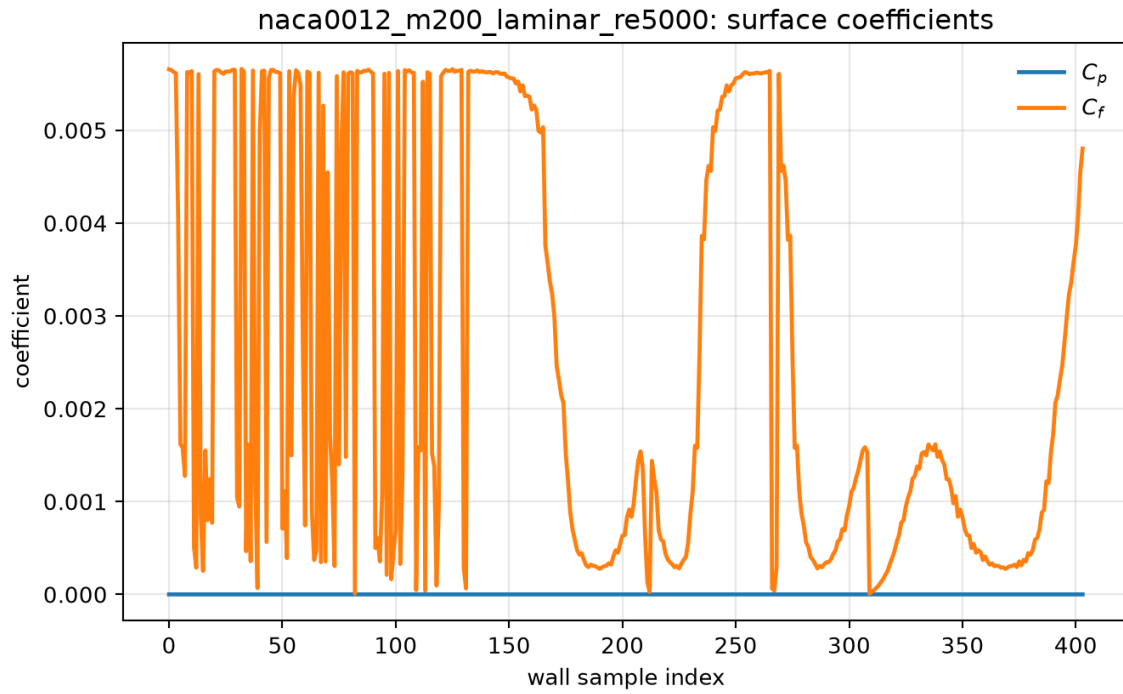


Figure 56: naca0012_m200_laminar_re5000: line plot of pressure (C_p) and skin-friction (C_f) surface coefficients (full domain) derived from surface.csv.

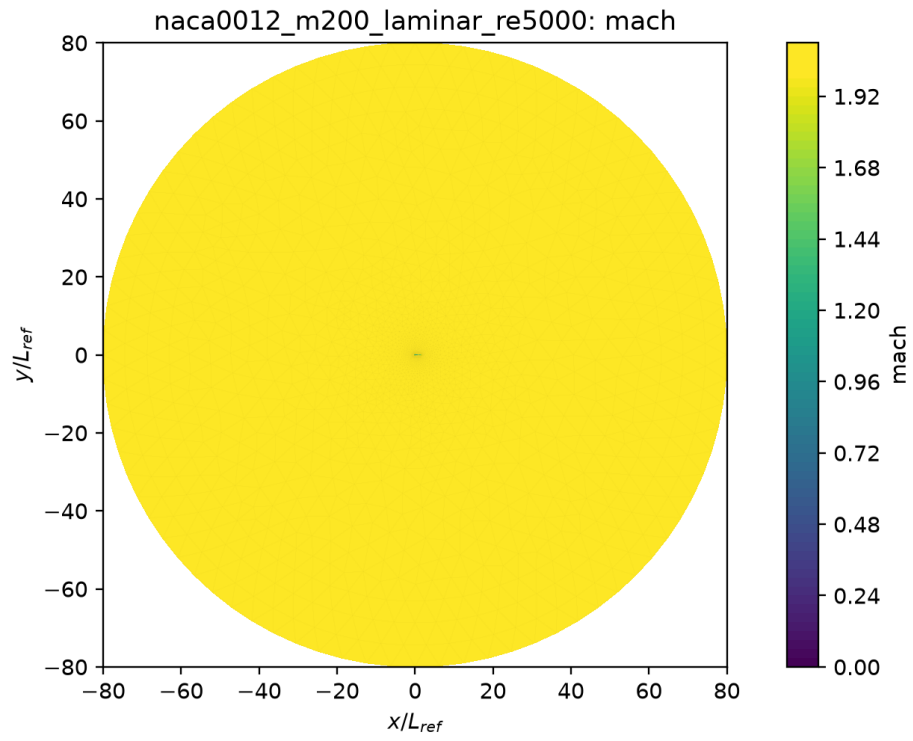


Figure 57: naca0012_m200_laminar_re5000: color-filled mach contour (full domain) derived from field_final.vtk.

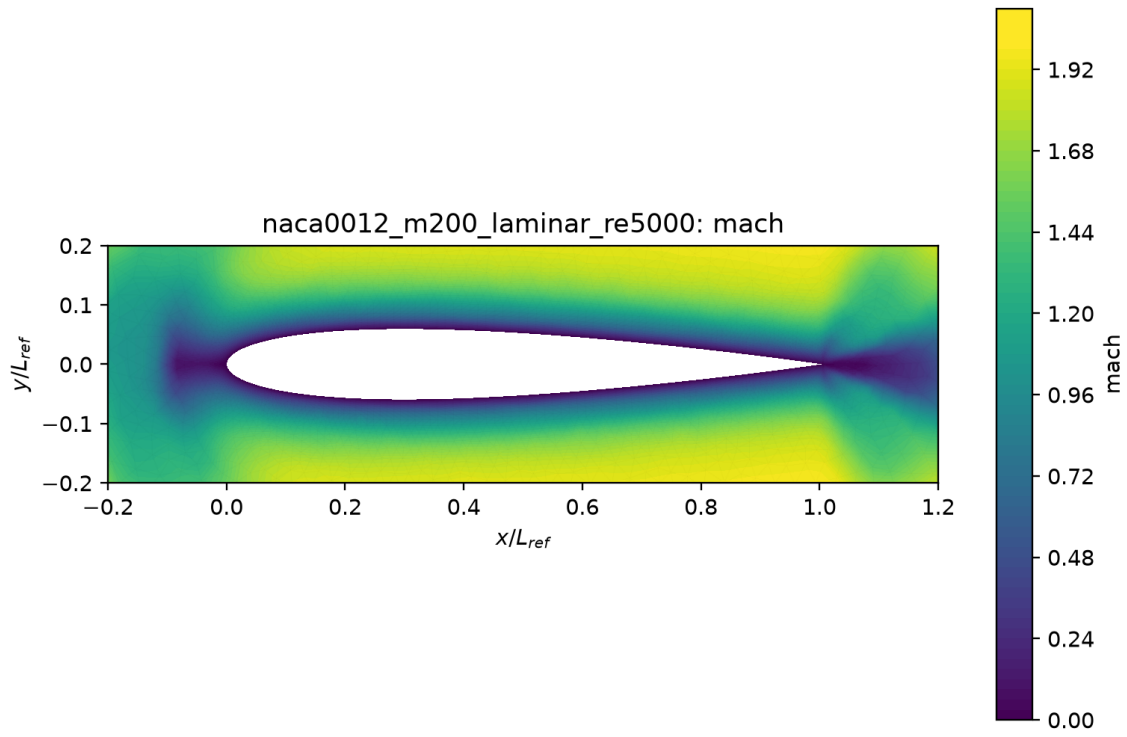


Figure 58: naca0012_m200_laminar_re5000: color-filled mach contour (body/wake zoom) derived from field_final.vtk.

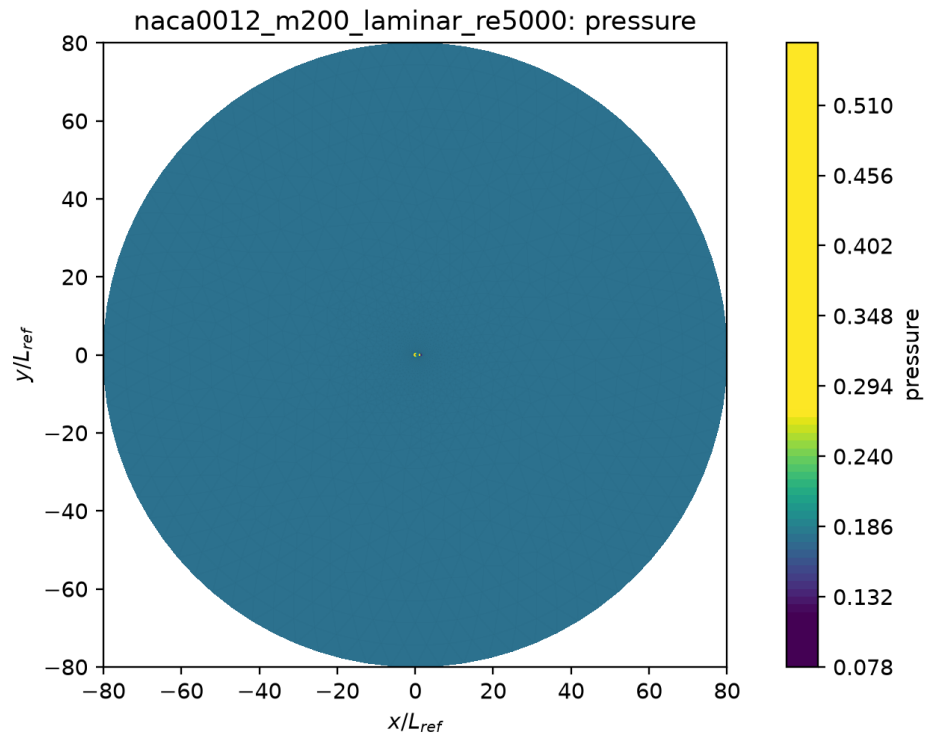


Figure 59: naca0012_m200_laminar_re5000: color-filled pressure contour (full domain) derived from field_final.vtk.

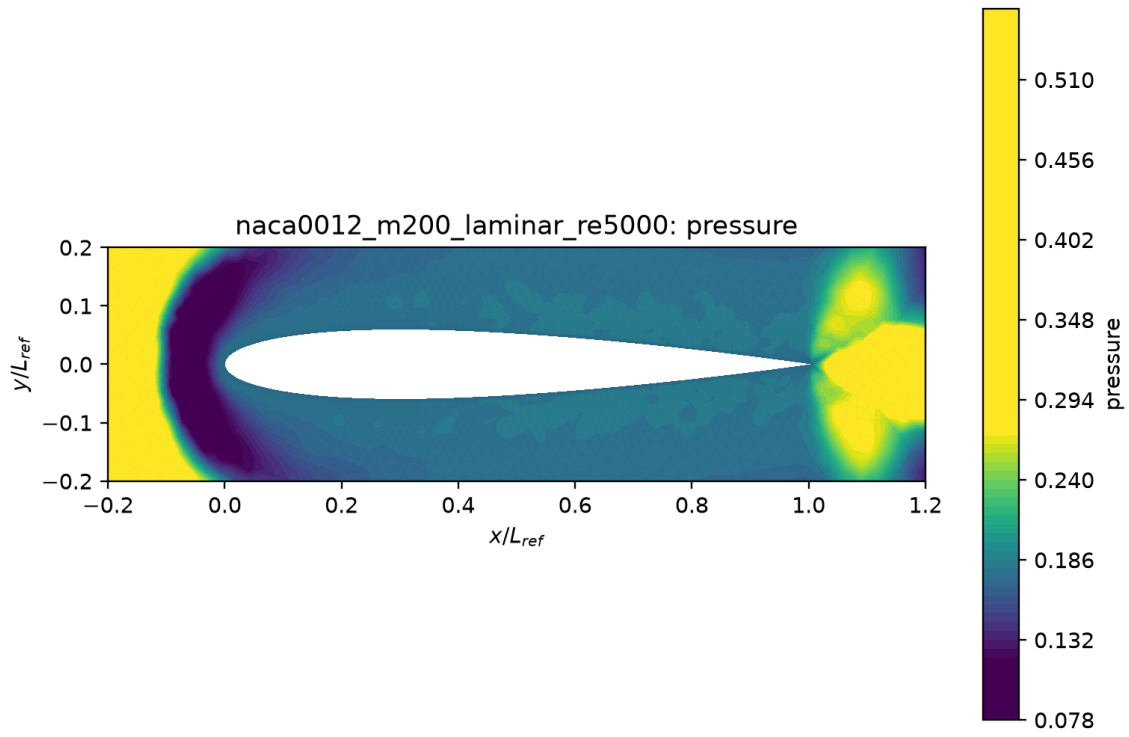


Figure 60: naca0012_m200_laminar_re5000: color-filled pressure contour (body/wake zoom) derived from field_final.vtk.

8 Limitations and reproducibility

The automated evidence identifies these limitations: `cylinder_m010_laminar_re20` is a stable force/residual plateau; the requested residual target was not reached; `cylinder_m010_laminar_re200` is explicitly marked failed because its terminal residual/force evidence did not meet the completion gate; the `Re=200` run does not demonstrate a resolved vortex-street lift signal; `naca0012_m015_inviscid` is a stable force/residual plateau; the requested residual target was not reached; `naca0012_m015_laminar_re5000` is a stable force/residual plateau; the requested residual target was not reached; `naca0012_m080_inviscid` is a stable force/residual plateau; the requested residual target was not reached; `naca0012_m080_laminar_re5000` is a stable force/residual plateau; the requested residual target was not reached; `naca0012_m200_inviscid` is a stable force/residual plateau; the requested residual target was not reached; `naca0012_m200_laminar_re5000` is a stable force/residual plateau; the requested residual target was not reached. The dominant remaining numerical limitation is the bounded scalar Jacobi/Richardson relaxation (with the optional block correction disabled for the canonical packages): some steady cases plateau before the requested residual orders, and the `Re=200` lift spectrum must be judged from the measured amplitude rather than assumed periodicity. Regenerate figures and this document with `tools/postprocess.py` and `tools/generate_report.py`; run commands, rank counts, wall times, and any launcher-observed return codes are in `run_manifest.csv`. A blank return code with source “not recorded” means the package was produced outside `run_cases.py`; it is not an inferred successful launch.